Finding a separation method in 25 injections: random search, Bayesian optimisation and an LLM chemist on three simulated mixtures

Three synthetic analyte sets stand in for unknown mixtures. A deterministic simulator turns any linear-gradient method into a retention table and a chromatographic response function (CRF) score, pre-computed on a 22 080-method grid per set. Three optimisers query that table one injection at a time with a budget of 25. This report shows what the samples and tables contain, how the CRF landscape looks, what happens when all three searches start from the same eight observations, how each search walks through the landscape from a cold start, and how often each of the three reaches the optimum over 50 repeated campaigns per set.

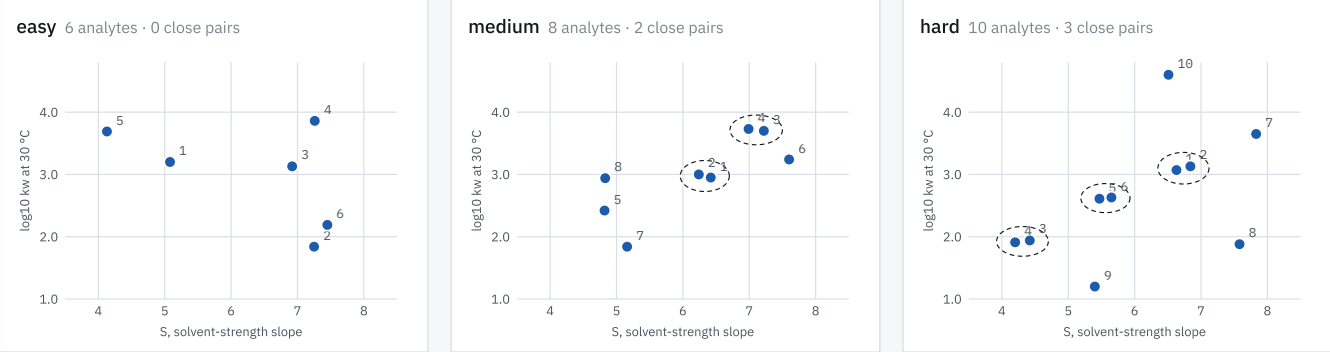
date 2026-10-21 budget 25 injections optimum CRF 1.000 on every set found = within 1 % of the optimum warm start on BO's 8 rows in section 6 + two real-compound sets in section 9 + the prompt taken apart in section 10 + a second model in section 11 + three fresh mixtures in section 12 updated 2026-11-07

4/50	20/50	50/50	0.4 %
Random search: campaigns reaching the optimum on the hard set · mean best @25 0.917	Bayesian optimisation: campaigns reaching the optimum on the hard set · mean best @25 0.975	LLM search: campaigns reaching the optimum on the hard set · mean best @25 1.000	Hard table: of 22 080 grid methods score CRF ≥ 0.99 (82 rows)

1. The three simulated samples

Each sample is a list of hidden analytes. Reverse-phase retention follows the linear-solvent-strength model with a van 't Hoff temperature term: for analyte *i*, $\log_{10} k = \log k_w - S \cdot \varphi + \Delta H\,(1/T_{ref} - 1/T)$. Three numbers per analyte therefore define a sample: $\log k_w$ (retention in pure water at 30 °C), S (how fast retention falls with organic fraction) and ΔH (how fast it falls with temperature). The optimiser never sees these; it only sees the retention table a method produces.

Difficulty is set by how many analytes are near-twins. Two analytes with close $\log k_w$ and close S co-elute over most of the method space and can only be pulled apart by a narrow choice of slope and temperature. The easy set has no such pair, the medium set two, the hard set three.



Hidden analyte parameters of each set, numbered by row in the config. Dashed rings mark clusters of closely eluting analytes ($|\Delta \log k_w| \le 0.3$ and $|\Delta S| \le 1.0$). All three stored optima reach CRF 1.0, so difficulty shows up as the size of the region a search must find, not as a lower ceiling.

easy · #	log kw	S	ΔH (K)
1	3.20	5.08	-2918
2	1.84	7.25	-1174
3	3.13	6.92	-1913
4	3.86	7.26	-2995
5	3.69	4.13	-1541
6	2.19	7.45	-1917

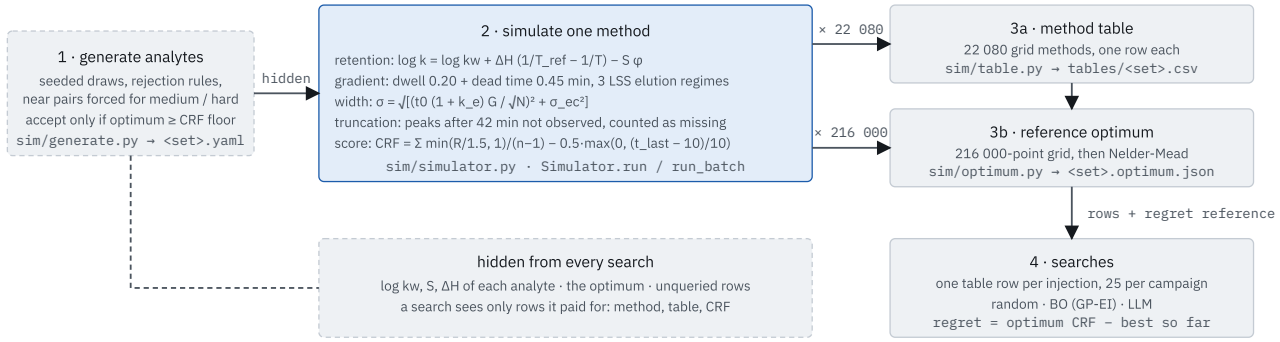
medium · #	log kw	S	ΔH (K)
1 close	2.95	6.42	-2058
2 close	3.00	6.24	-2530
3 close	3.70	7.22	-2503
4 close	3.73	6.99	-1352
5	2.42	4.82	-1750
6	3.24	7.60	-1050
7	1.84	5.16	-2238
8	2.94	4.83	-2402

hard · #	log kw	S	ΔH (K)
1 close	3.07	6.63	-2946
2 close	3.13	6.84	-1592
3 close	1.94	4.42	-1914
4 close	1.91	4.20	-1362
5 close	2.61	5.47	-2873
6 close	2.63	5.65	-1905
7	3.65	7.83	-1152
8	1.88	7.58	-2097
9	1.20	5.40	-1974
10	4.60	6.51	-2336

set	analytes	optimum φ0	φ1	tG (min)	T (°C)	CRF	last peak (min)	rows with CRF ≥ 0.99
easy	6	0.05	0.536	3.31	50	1.000	9.68	1 112 of 22 080
medium	8	0.05	0.398	2.00	25	1.000	8.52	500 of 22 080
hard	10	0.05	0.481	5.93	55	1.000	9.11	82 of 22 080

2. How the data was prepared

Everything a search sees comes out of one deterministic pipeline: a seeded generator draws the hidden analytes, a simulator turns any method into a retention table and a score, the simulator is evaluated on a round-number grid to make the tables, and a dense grid plus local refinement finds the reference optimum each search is scored against. No measurement noise is added anywhere; two runs of the same method give the same row.



The pipeline from hidden analytes to the rows a search can buy. The simulator is the same function at every stage; the table is simply the simulator evaluated on the grid, and the stored optimum is the simulator maximised off-grid.

Step 1 · Generating the samples

Analytes are drawn by rejection sampling from uniform ranges, and a draw is kept only if it differs from every accepted analyte by more than 0.3 in $\log k_w$ or more than 1.0 in S. Difficulty is then imposed by overwriting rows into near pairs, and a set is accepted only if its reference optimum reaches a CRF floor. The sets used here are seeds 0, 1000 and 2000 of that procedure.

set	analytes	construction rule	CRF floor	seed
easy	6	$\log k_w$ ~ U[1.8, 4.0], S ~ U[4, 8], ΔH ~ U[−3000, −1000]; all pairwise distinct	0.9	0
medium	8	easy rule, then two pairs overwritten to within 0.08 in $\log k_w$ and 0.15 to 0.30 in S (equal S would be inseparable at every φ)	0.9	1000
hard	10	medium rule with three near pairs, plus one very early analyte ($\log k_w$ = 1.2) and one very late one ($\log k_w$ = 4.6)	0.7	2000

Step 2 · Simulating a method

Retention. Isocratic retention is the linear-solvent-strength relation with a van 't Hoff temperature term, $\log_{10} k = \log k_w + \Delta H\,(1/T_{ref} - 1/T) - S\,\varphi$, with T in kelvin and T_{ref} = 30 °C. S is temperature-independent. Two analytes whose log k lines cross swap elution order at $\varphi = \Delta \log k_w / \Delta S$, which is why the near pairs respond to the φ window and not only to slope.

Gradient elution. The program is linear from φ_0 to φ_1 over t_G, then held at φ_1. It reaches the column inlet 0.20 min late (dwell) and the outlet a further 0.45 min later (dead time). The analyte's migration integral is solved in closed form in three regimes: eluted before the gradient front arrives, eluted during the gradient (the textbook LSS case, with steepness $b = t_0\,S\,\Delta\varphi / t_G$), or still on the column when the gradient ends and finishing in the hold at φ_1. The closed form is checked against a Runge-Kutta integration of the same model in the test suite.

Peak width. $\sigma = \sqrt{[(t_0\,(1 + k_e)\,G / \sqrt{N})^2 + \sigma_{ec}^2]}$ with N = 15 000 plates, k_e the retention factor at the moment of elution, $G = 1 - 0.25\,b / (1 + b)$ the gradient band-compression factor for

a peak eluting inside the gradient (G = 1 otherwise) and σ_{ec} = 0.02 min of extra-column dispersion, so no peak is narrower than that.

Truncation. Acquisition stops at 42 min, which is the 45 min maximum run minus 3 min re-equilibration. Peaks after that are dropped from the table and counted as missing. The spec made such runs infeasible; the implementation keeps them feasible with a poor score instead, because an operator does see a truncated chromatogram, and a cliff from −1.6 to "no data" at 42.0 min would have cut an otherwise smooth landscape.

Score. CRF = [Σ over adjacent observed pairs of min(R / 1.5, 1)] / (n − 1) − 0.5 · max(0, (t_{last} − 10) / 10), with R = Δt_R / (2 (σ_i + σ_{i+1})). A missing analyte is one pair with zero credit and sets t_{last} to 42 min, so the floor is −1.6. The maximum of 1.0 needs every pair at R ≥ 1.5 and the last peak within 10 min: the best method is the shortest fully resolved one. Pressure was dropped from the study because the 600 bar limit never binds on this grid (worst case about 530 bar).

A worked example: one method through the simulator

The hard set at the LLM search's best method from section 8 (φ 0.05 → 0.80 over 10 min at 55 °C), with the intermediate quantities an operator never sees. Rows are in elution order; the analyte number refers to the config. All ten elute inside the gradient (regime 2), the two members of each near pair sit next to each other, and the late analyte 10 arrives last at 7.6 min, inside the 10 min window.

analyte	log kw	S	ΔH	log kw at 55 °C	k at φ0	b	regime	k_e	t_R (min)	σ (min)	R to next
9	1.20	5.40	-1974	0.70	2.7	0.182	2	1.39	1.368	0.0217	7.92
8	1.88	7.58	-2096	1.35	9.4	0.256	2	1.50	2.057	0.0218	12.04
3	1.94	4.42	-1913	1.46	17.2	0.149	2	2.54	3.150	0.0236	4.45
4	1.91	4.20	-1362	1.57	22.9	0.142	2	2.75	3.575	0.0241	**1.59**
5	2.61	5.47	-2872	1.89	41.0	0.185	2	2.25	3.725	0.0231	4.07
1	3.07	6.63	-2945	2.32	98.6	0.224	2	1.91	4.096	0.0225	**1.52**
6	2.63	5.65	-1904	2.15	73.3	0.191	2	2.22	4.234	0.0230	5.98
2	3.13	6.84	-1592	2.73	244.4	0.231	2	1.87	4.777	0.0224	6.32
7	3.65	7.83	-1151	3.36	931.2	0.265	2	1.64	5.338	0.0220	25.38
10	4.60	6.51	-2336	4.01	4.9e+3	0.220	2	1.98	7.602	0.0226	–

CRF 1.000 · run time 10.6 min including re-equilibration · regime 2 = eluted during the gradient · b = t0 · S · Δφ / tG

Step 3 · The table and the reference optimum

The table is the simulator evaluated on every in-bounds point of a round-number grid (φ in steps of 0.05, t_G in steps of 2 min, T in steps of 5 °C), which is what the next section lists column by column. The reference optimum is found off-grid: a 30 × 30 × 30 × 8 grid over (φ_0, Δφ, t_G, T), 216 000 points evaluated in batch, then Nelder-Mead from the best point. A search is scored against that CRF, and "found" means within 1 % of it. On all three sets the optimum is 1.0, and the best grid row also reaches 1.0, so the discrete table loses nothing at the top.

3. What is in the simulated table

For each set, `tables/<set>.csv` holds one row per method on a round-number grid inside the instrument bounds: φ_0 0.05 to 0.60 and φ_1 0.15 to 0.95 in steps of 0.05 (with $\varphi_1 \geq \varphi_0 + 0.10$), t_G 2 to 40 min in steps of 2, and T 25 to 60 °C in steps of 5. That is 12 × 17 × 20 × 8 minus the φ_1 constraint, or 22 080 methods per set. The row is everything the instrument would report for that injection and nothing about the analytes.

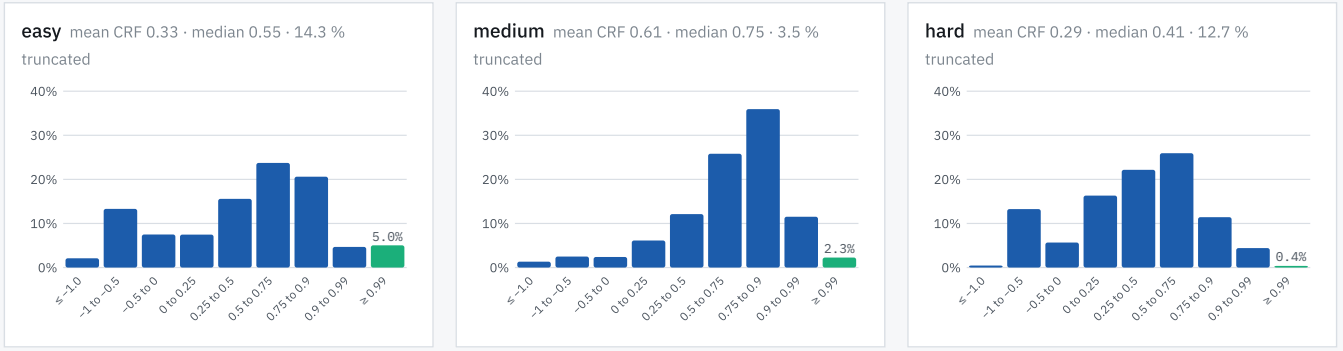
The instrument is a 100 × 2.1 mm column with 1.8 µm particles, N = 15 000 plates, 0.5 mL/min, dead time 0.45 min, dwell 0.20 min. Acquisition stops at 42 min (45 min maximum minus 3 min re-equilibration); analytes still on the column then are not observed, the row stays feasible with a short table and a poor score, and `n_missing` counts them. No row of the default grid is infeasible.

column	meaning
`method_id`	row index, stable for a given grid
`phi0, phi1, tG, T`	the method: initial and final organic fraction, gradient time (min), column temperature (°C)
`feasible, reason`	False only for a method outside the bounds (never on the default grid)
`crf`	the score: resolution term minus time term, in [−1.6, 1]
`resolution_term`	Σ over adjacent observed pairs of min(R / 1.5, 1), divided by n − 1; a missing analyte is one pair with zero credit
`time_term`	$0.5 \cdot \max(0, (t_{last} - 10) / 10)$, with $t_{last} = 42$ whenever a peak is missing
`min_resolution, n_unresolved_pairs`	worst adjacent resolution and the number of observed pairs with R < 1.5
`t_first, t_last, run_time`	first and last observed retention time; $\min(t_{last}, 42)$ + 3 min re-equilibration
`n_missing, n_observed`	analytes not eluted by 42 min, and rows in the retention table
`retention_table`	JSON list of [t_R, σ] for the observed peaks sorted by t_R, no identities

Six rows of the hard table

method_id	phi0	phi1	tG	T	crf	res. term	time term	min R	t_last	missing	retention_table (t_R, σ)
0	0.05	0.15	2	25	-1.047	0.553	1.600	1.47	41.28	4	[2.87, 0.025] [4.93, 0.036] [11.56, 0.090
5000	0.1	0.9	12	25	0.773	0.773	0.000	0.21	9.25	0	[1.99, 0.023] [2.62, 0.022] [4.43, 0.025]
11029	0.25	0.65	38	50	0.371	0.902	0.530	0.45	20.60	0	[0.58, 0.021] [0.61, 0.021] [1.64, 0.024]
13311	0.3	0.8	8	60	0.765	0.765	0.000	0.01	4.56	0	[0.49, 0.020] [0.49, 0.020] [0.92, 0.021]
9478	0.2	0.85	10	55	0.881	0.881	0.000	0.00	6.22	0	[0.64, 0.021] [0.75, 0.021] [1.68, 0.022]
21653	0.6	0.85	14	50	0.164	0.164	0.000	0.01	1.16	0	[0.45, 0.020] [0.45, 0.020] [0.46, 0.020]

How the scores are distributed



Share of the 22 080 grid methods per CRF bucket. The right-most bucket (CRF ≥ 0.99, counted as the optimum) holds 1 112 rows on easy, 500 on medium and 82 on hard: 5.0 %, 2.3 % and 0.4 % of the table. A method drawn at random lands there once in 20, 44 or 269 draws respectively.

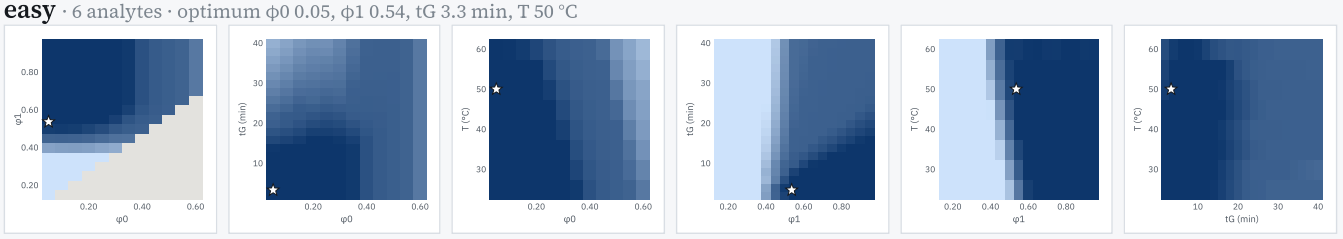
set	rows	feasible	truncated (peaks missing)	mean CRF	median CRF	min CRF	rows ≥ 0.9	rows ≥ 0.99	best grid CRF
easy	22 080	100 %	14.3 %	0.333	0.551	-1.40	2145	1112	1.0
medium	22 080	100 %	3.5 %	0.609	0.749	-1.60	3044	500	1.0
hard	22 080	100 %	12.7 %	0.293	0.406	-1.34	1056	82	1.0

4. CRF landscape of the simulated tables

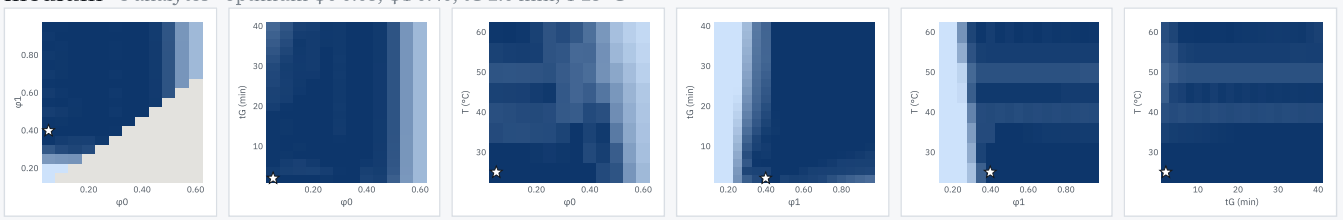
The method space has four dimensions, so each set is drawn as six pairwise panels. A cell shows the best CRF over the two variables not on the axes (a max-projection: "is there any good method with these two values?"). Grey cells contain no method because φ_1 must exceed φ_0 by 0.10. The star is the stored continuous optimum found by a 216 000-point grid plus Nelder-Mead.

The projection is deliberately optimistic: a dark cell only says that some method with those two values is good. On the hard set the dark region in the φ_0 × T panel is wide, yet only 82 of 22 080 methods actually score 0.99 or better, because the two near pairs also need the right slope.

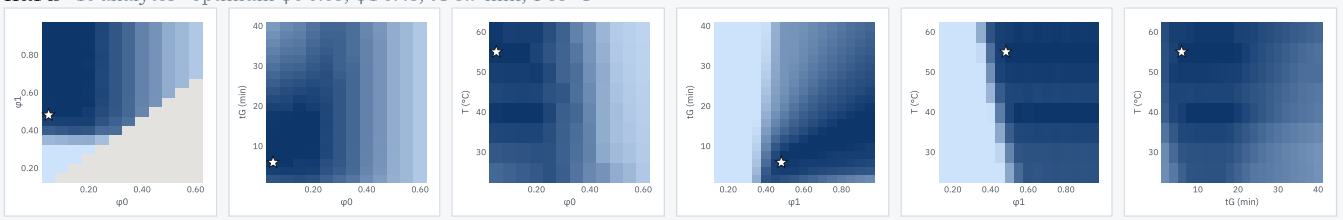
easy · 6 analytes · optimum ϕ0 0.05, ϕ1 0.54, tG 3.3 min, T 50 °C



medium · 8 analytes · optimum ϕ0 0.05, ϕ1 0.40, tG 2.0 min, T 25 °C



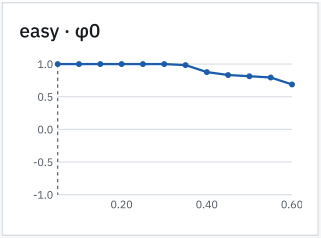
hard · 10 analytes · optimum ϕ0 0.05, ϕ1 0.48, tG 5.9 min, T 55 °C

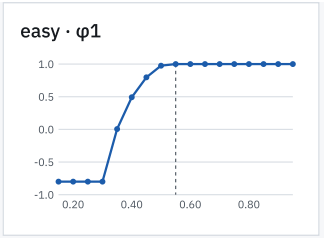


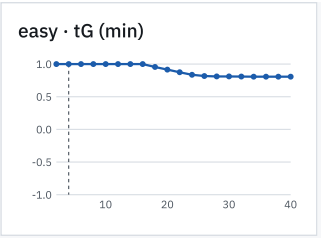
CRF 0 → 1 (values below 0 drawn as 0) no method ($\varphi_1 < \varphi_0 + 0.10$) ★ stored optimum

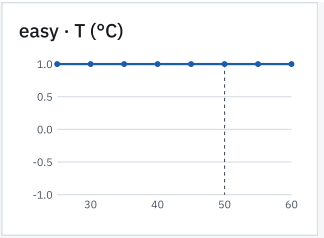
Best CRF against each variable alone

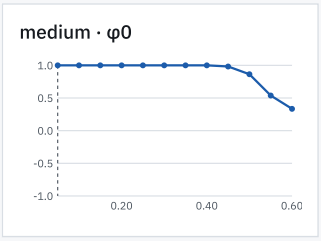
The diagonal of the corner plot: the best CRF over the other three variables as a function of one. A flat top means the variable is not what limits the set; a narrow peak means it is. On the hard set only two temperatures, 40 and 55 °C, hold any method scoring 0.99 or better, which is the single fact that makes it hard.

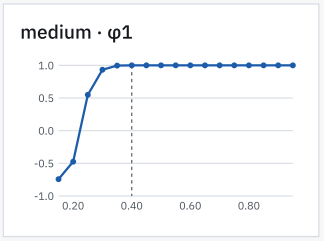


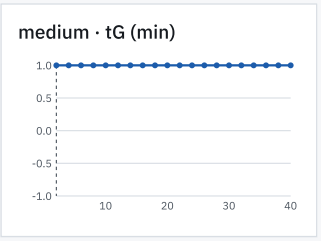


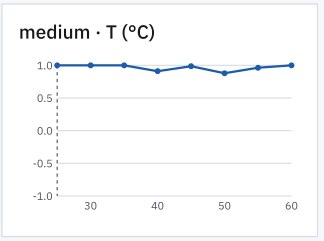


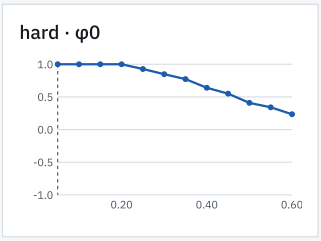


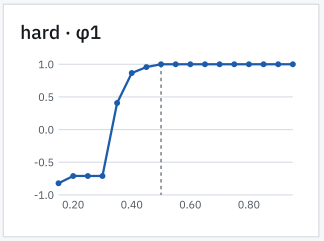


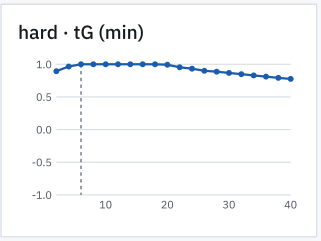


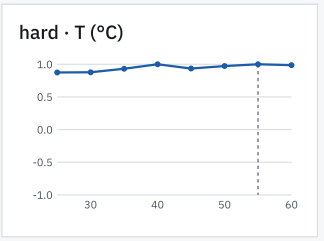












Where the optimal methods are

Extent of the region of grid methods with CRF ≥ 0.99 in each variable, and how its rows are spread over temperature. Easy is a broad plateau at short gradients; medium is cold (25 to 35 °C) and tolerant of any gradient length; hard is a thin ridge that needs a low φ_0, a 6 to 20 min gradient and one of two temperatures.

set	rows with CRF ≥ 0.99	φ0	φ1	tG (min)	T (°C)	optimal rows by temperature
easy	1112	0.05 to 0.30	0.55 to 0.95	2 to 16	25 to 60	25 °C: 172 · 30 °C: 189 · 35 °C: 193 · 40 °C: 160 · 45 °C: 163 · 50 °C: 122 · 55 °C: 104 · 60 °C: 9
medium	500	0.05 to 0.40	0.35 to 0.95	2 to 40	25 to 60	25 °C: 318 · 30 °C: 138 · 35 °C: 30 · 60 °C: 14
hard	82	0.05 to 0.20	0.50 to 0.95	6 to 20	40 to 55	40 °C: 24 · 55 °C: 58

5. The three searches and the LLM harness

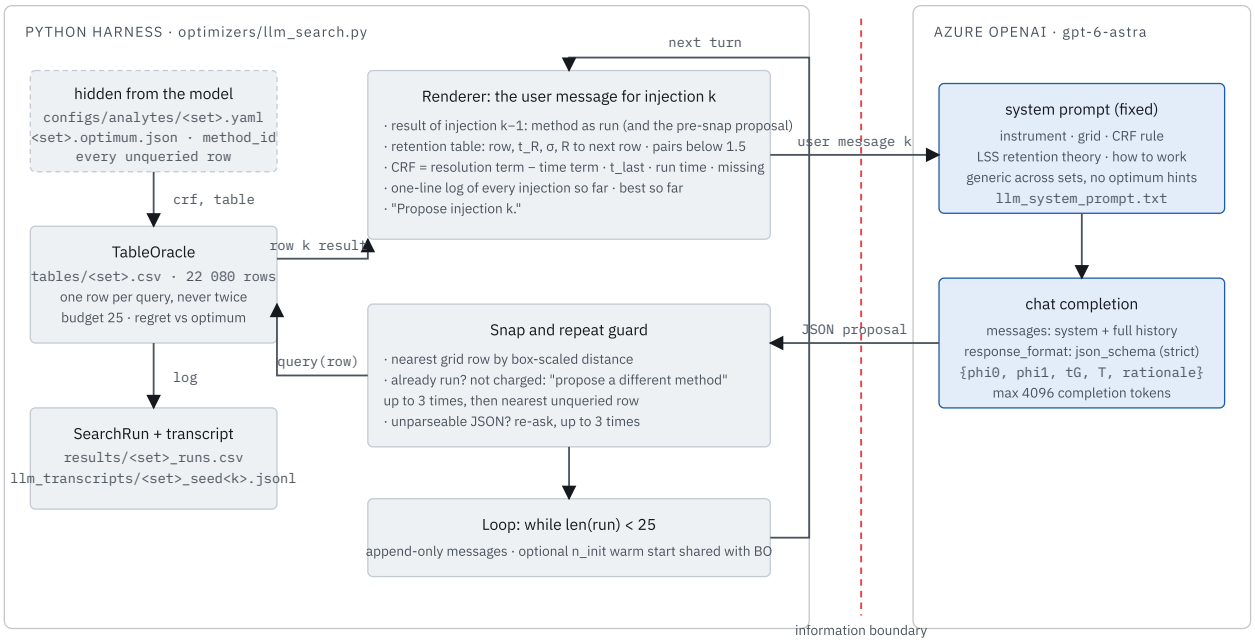
All three optimisers share one contract: a `TableOracle` over the set's table, a budget of 25, one row per query, a row never charged twice, regret measured against the stored optimum. Random search draws a seeded permutation of the rows. Bayesian optimisation (BO) uses the same first eight random rows, then fits a Gaussian process (ARD Matérn-5/2, hyperparameters refit each step) and picks the unqueried row with the highest expected improvement. The LLM search starts from a blank page: an Azure OpenAI deployment (`gpt-6-astra`) reads the retention table of every injection so far and proposes the next method as JSON, which the harness snaps to the grid.

method	what it knows before injecting	how it chooses the next row	first eight injections	cost per campaign
Random search	nothing; it never reads a result	the next row of a seeded permutation of all 22 080	random rows	milliseconds
Bayesian optimisation	only the CRF of each row it has run; the retention table is not used	fits a Gaussian process (ARD Matérn-5/2, hyperparameters refit each step, lognormal prior on lengthscales) on the unit cube and takes the unqueried row with the highest expected improvement	the same random rows as random search for that seed	about 10 s of CPU
LLM search	the full retention table of every injection with pairwise resolution, the CRF and its terms, plus the instrument, grid, scoring rule and LSS theory in its prompt	proposes one method as JSON with a rationale; the harness snaps it to the nearest grid row and refuses repeats	its own: a broad scouting gradient, then one-step changes	25 API calls, about 290 000 prompt tokens, 3 to 5 min

Random search and BO need no more description than the table gives. The rest of this section is the LLM search: how the harness turns a proposal into an injection, what the system prompt says, and what the model is shown after each injection.

Architecture

The Python harness owns the budget, the grid, the logging and the conversation. The model answers one question per injection: given everything so far, which method next? Nothing from the analyte parameters, the optimum or any unqueried table row crosses the dashed boundary.



One chat call per injection. The harness renders what the instrument would show an operator (retention table, pairwise resolution, CRF and its terms) into the user message; the model returns one method as strict JSON; the harness snaps it to the grid, refuses repeats without charging them, queries the table and writes the transcript. The analyte parameters, the stored optimum and all unqueried rows stay left of the dashed line.

set	campaigns	API calls	prompt tokens per campaign	completion tokens per campaign	mean latency per call (s)	proposals re-asked (repeat of a run method)	unparseable replies
easy	50	1,250	268,644	2,906	4.6	0	0
medium	50	1,250	283,667	6,620	8.7	0	0
hard	50	1,250	295,053	6,644	9.3	0	0

System prompt

The text below is `optimizers/llm_system_prompt.txt` verbatim, the fixed first message of every conversation. It gives the instrument, the grid, the scoring rule and standard linear-solvent-strength method-development practice. It is generic across sets: it does not state the number of analytes (the first user message does) and carries no hint about where optima tend to lie.

```
You are developing a reverse-phase HPLC gradient method for a mixture whose
composition you do not know. You choose the method; the instrument runs it and
returns the chromatographic result. Your goal is the highest CRF score reachable
within a fixed number of injections.

## Instrument (fixed for the campaign)
- Column 100 x 2.1 mm, 1.8 um particles, N = 15 000 plates. Flow fixed at 0.5 mL/min.
- Dead time t0 = 0.45 min. Dwell (gradient delay) tD = 0.20 min: the programmed
  gradient reaches the column inlet at 0.20 min and the outlet 0.45 min after that,
  so at the outlet the gradient runs from 0.65 min to tG + 0.65 min.
- Extra-column dispersion sigma_ec = 0.02 min; no peak is narrower than that.
- Data acquisition stops at 42 min (45 min maximum run minus 3 min
  re-equilibration). Compounds that have not eluted by then are not observed:
  the retention table is shorter and you are told how many are missing.

## Method (what you choose)
A linear gradient from phi0 to phi1 (organic fraction) over tG minutes, then a hold
at phi1 until the run ends, at column temperature T.
The instrument only runs grid values; anything else is rounded to the nearest one:
- phi0: 0.05 to 0.60 in steps of 0.05
- phi1: 0.15 to 0.95 in steps of 0.05, and phi1 >= phi0 + 0.10
- tG:   2 to 40 min in steps of 2
- T:    25 to 60 C in steps of 5

## What a run returns
The retention table: one row per observed peak sorted by retention time, with
t_R (min), sigma (peak standard deviation, min) and the resolution to the next
row, R = (t_R,next - t_R) / (2 (sigma + sigma_next)); plus the CRF and its two
terms. The mixture has a known number of compounds (stated in the first
message). When all of them elute within 42 min the table has exactly that many
rows: co-elution shows up as R near 0, not as a missing row. Rows carry no
identities. The same compound can be row 3 in one run and row 5 in another when
the elution order changes, so compare runs by pattern, not by row number.

## The score
CRF = [sum over adjacent observed pairs of min(R / 1.5, 1)] / (n - 1)
      - 0.5 * max(0, (t_last - 10) / 10)
with n the number of compounds in the mixture.
- A pair earns full credit at R >= 1.5 and nothing beyond it, so the score is lost
  at the worst pairs.
- A last peak after 10 min costs 0.05 per extra minute.
- A compound missing at 42 min counts as one pair with zero credit, and t_last is
  then taken as 42 min, so a truncated run pays 1.6 on the time term alone.
  The floor is -1.6.
- 1.0 is the maximum: every pair at R >= 1.5 with the last peak within 10 min.
  The best method is therefore the shortest fully resolved one.

## Retention model (what the chemistry lets you infer)
Reverse-phase retention follows the linear-solvent-strength model: for each
compound log10 k = log kw - S * phi at a given temperature, with compound-specific
log kw (roughly 1.5 to 4.5) and S (roughly 3 to 10). Raising temperature lowers
retention for every compound by a compound-specific amount, so T shifts
selectivity modestly (it can reorder a close pair) and shortens the run.
Consequences you can use:
- The organic fraction at which a peak leaves the column is about
  phi_e = phi0 + (phi1 - phi0) * (t_R - 0.65) / tG, clamped to [phi0, phi1].
  A peak at or before about 0.6 min is essentially unretained: lower phi0.
  A peak after tG + 0.65 min eluted during the hold at phi1: raise phi1 or lengthen tG.
- Trimming the gradient window to the elution window (phi0 a little below phi_e of
  the first peak, phi1 a little above phi_e of the last) keeps every peak inside
  the gradient and spends the run time where the peaks are.
- Gradient steepness b = t0 * S * (phi1 - phi0) / tG sets resolution. Shallower
  (longer tG, narrower window) spreads pairs whose S differ and widens peaks a
  little; steeper compresses and shortens. A pair whose members have nearly equal
  S barely responds to slope and needs T or a different phi window instead.
- Two runs that differ only in tG show which pairs respond to slope (their R
  changes a lot) and which do not.

## How to work
- Propose exactly one method per injection. Each injection spends one unit of
  budget; a proposal that rounds to a method already run is wasted, so stay on the
  grid and do not repeat.
- Early injections buy information: where the elution window is, which pairs are
  critical, how they respond to slope and temperature. Later injections exploit it.
  Keep the best method in mind and refine around it as the budget runs out.
- Once every pair is at R >= 1.5, the remaining gain is time: shorten tG, narrow the
  window or raise T until a pair starts to fall below 1.5, then back off one step.
```

```
- Avoid truncated runs: a shallow gradient ending at a low phi1 can hold the most
  retained compound on the column past 42 min.

Answer with JSON only:
{"phi0": <number>, "phi1": <number>, "tG": <number>, "T": <number>,
 "rationale": "<one or two sentences>"}
```

Per-injection user message

Rendered by the harness after every injection; the only text that varies. Shown here for the hard set, seed 0: the opening message, then the message after injection 1.

```
No injections yet. The mixture contains 10 compounds. You have 25 injections.

Propose injection 1.

--- after injection 1 ---

Injection 1 of 25 run, 24 remaining. Best so far: CRF 0.918 at injection 1.

Result of injection 1: phi0 0.05, phi1 0.95, tG 10 min, T 40 C  (as proposed)
observed peaks: 10 of 10
CRF 0.918 = resolution term 0.918 - time term 0.000
t_last 7.16 min (last observed peak), run time 10.2 min
peaks (sorted by t_R):
 row     t_R   sigma   R_to_next
   1    1.75  0.0219      6.75
   2    2.34  0.0216     13.56
   3    3.55  0.0231      2.28
   4    3.76  0.0234      4.72
   5    4.20  0.0225      1.44
   6    4.33  0.0224      0.46
   7    4.37  0.0220      2.57
   8    4.59  0.0219      3.61
   9    4.91  0.0216     25.82
  10    7.16  0.0220         -
pairs below 1.5: (5,6) R 1.44, (6,7) R 0.46

All injections so far:
  #   phi0  phi1   tG   T   seen    CRF   minR  t_last  worst pair
  1   0.05  0.95   10  40  10/10   0.918   0.46     7.2  (6,7)

Propose injection 2.
```

Reading the result

In the seed-0 campaigns of section 7 the LLM search finds a CRF 1.0 method on every set within five injections and spends the rest of the budget looking for faster or more robust methods at the same score; section 8 shows how that holds up over 50 campaigns. Two things make the behaviour plausible rather than lucky. The prompt tells the model that the best method is the shortest fully resolved one and how slope, window and temperature act on resolution, so its first scouting gradient already lands in the dark region of the landscape, and each later move is one grid step with a stated reason. And the hard set's difficulty is that the good region is small, not that it is hidden: a reader who knows the chemistry can walk to it.

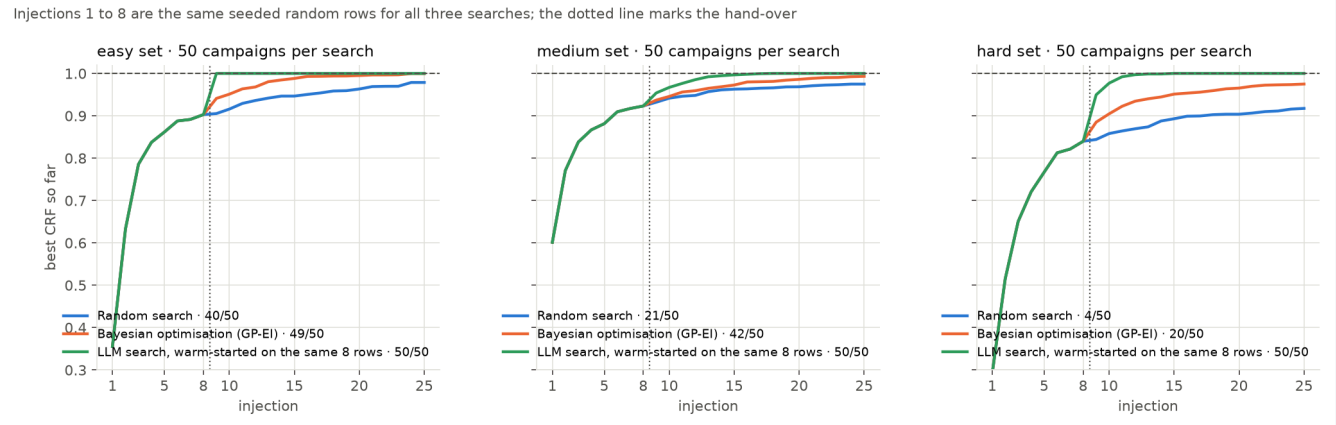
With 50 campaigns per method the comparison is symmetric in budget and repeats; it is not symmetric in information, since the LLM search carries a domain prior in its prompt that neither baseline has. The checks that measure that prior are a warm-started variant sharing BO's first eight injections (section 6), the prompt taken apart piece by piece (section 10), the same prompt and harness driven by a second model (section 11) and a memorisation check on freshly generated analyte sets (section 12).

6. Same observations, different readers: the warm-start comparison

The cold-start campaigns of sections 7 and 8 let the LLM search choose its own scouting runs, so two things differ between it and BO: what each reads and where each starts. This section removes the second difference, and it is the comparison the rest of the report is built on. Every LLM campaign was started on the same eight seeded random injections that open the BO and random-search campaigns of the same seed, with the eight retention tables and scores placed in its conversation as if it had proposed them, and it then proposed the remaining 17 injections itself. Same model (`gpt-6-astra`), prompt, harness, grid and budget as everywhere else, 50 campaigns per set. The three searches therefore hold the same observations at injection 8 and diverge only in what they make of them: random search draws on, BO fits a Gaussian process to eight scores, the LLM reads eight tables.

set	search	found optimum	injections to optimum, median (IQR; max)	of which its own, median (IQR; max)	found by 10 / 15	mean best CRF @25
easy	random	40 of 50	10 (5–17; 24)	–	23 / 29	0.979
	BO	49 of 50	9 (5–13; 23)	–	31 / 43	1.000
	LLM, warm-started	**50 of 50**	9 (5–9; 9)	1 (1–1; 2)	50 / 50	1.000
medium	random	21 of 50	13 (9–17; 24)	–	8 / 15	0.975
	BO	42 of 50	15 (11–19; 25)	–	10 / 24	0.993
	LLM, warm-started	**50 of 50**	11.5 (10–14; 18)	3.5 (2–6; 10)	17 / 41	1.000
hard	random	4 of 50	7 (6–10; 16)	–	3 / 3	0.917
	BO	20 of 50	14 (10–17; 25)	–	6 / 13	0.975
	LLM, warm-started	**50 of 50**	10 (10–11; 15)	2 (2–3; 7)	28 / 50	1.000

50 campaigns of 25 injections per search and set, all three starting from the same eight seeded random rows. "Found" is a best CRF within 1 % of the optimum; the injection statistics are over the campaigns that found it and count all 25 injections; for the LLM search the next column counts only its own injections after the eight shared ones. The random-search and BO rows are those of section 8, which share the same start. The eight rows contained the optimum in only 19, 5 and 3 campaigns of 50 and included 137 truncated runs across the three sets; the LLM's own 17 injections included none.



Mean best CRF so far against injection, 50 campaigns per search and synthetic set, all three searches starting from the same eight seeded random rows (dotted line: the hand-over after injection 8). The legend gives the campaigns that reached the optimum. File: `ablations/warm_start/results/warm_start.png` .

What each search made of the eight rows

Given the same eight random injections that open BO's campaign, the LLM reached the optimum in every campaign after a median of 1, 3.5 and 2 further injections of its own (at most 2, 10 and 7), where BO from the same rows reached it in 49, 42 and 20 campaigns after a median of 9, 15 and 14 injections in all, and random search in 40, 21 and 4. By injection 15 the LLM was at the optimum in 50, 41 and 50 campaigns, BO in 43, 24 and 13. Counting all 25, the warm-started LLM needed 9, 11.5 and 10, more than from a cold start

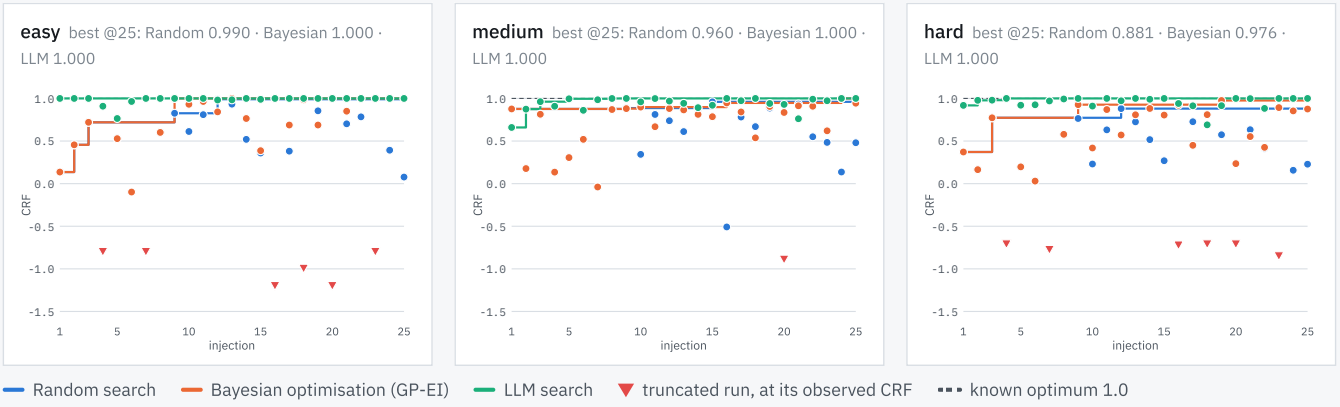
(1, 6 and 4; section 8), because eight random rows are worse than its own scouting gradient: they contained the optimum in only 19, 5 and 3 campaigns of 50 and included 137 truncated runs, which the LLM never proposes. The rows were not wasted on it, though. Its first own proposal after reading them scored 0.997, 0.929 and 0.944, against 1.000, 0.592 and 0.931 from a cold start, and on the medium and hard sets it then needed about two fewer injections of its own. The same eight observations carry BO to the optimum in 40 % of hard-set campaigns and the LLM in all of them.

> What remains asymmetric after the warm start is not the observations but what each search is told they mean: the LLM's prompt states the scoring rule and the instrument bounds, and the per-injection message gives it the two terms of the CRF. Section 10 shows how little of that it needs, and section 11 how much depends on the model behind the prompt. Files: `ablations/warm_start/` ; summarise with `python ablations/summary.py` , plot with `python ablations/warm_start/plot_warm_start.py` .

7. How each search navigates the landscape

Back to the cold start, where each search opens as it likes: random search and BO with their eight random rows, the LLM search with a scouting gradient of its own choosing. This section follows one campaign of each.

The figures below show seed 0 of each method on each set: the CRF of every injection as a dot, the best CRF so far as a step line. A truncated run (peaks still on the column at 42 min) is a red triangle at its observed score; it is an ordinary poor injection, not a failure.

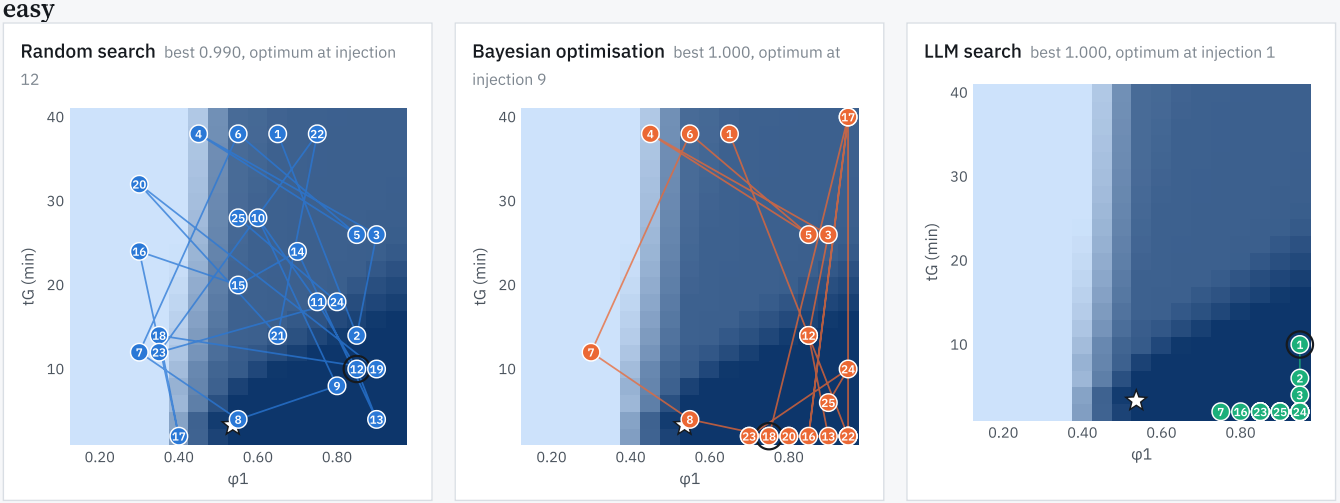


Seed 0, 25 injections. Random search and BO share injections 1 to 8. The LLM search reaches the optimum at injection 1 (easy), 5 (medium) and 4 (hard) and stays there; BO reaches it at injection 9 on easy and gets to 0.976 on hard; random search stalls at 0.990, 0.887 and 0.881.

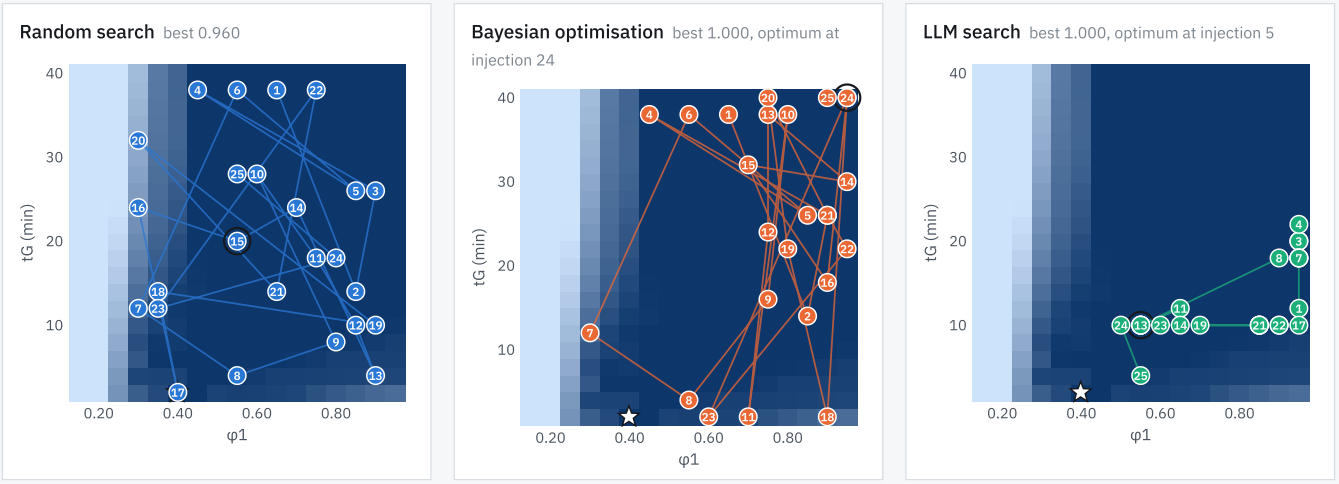
Where each search walked

The same seed-0 campaigns drawn on the $\varphi_1 \times t_G$ panel of the landscape (best CRF over φ_0 and T). Markers are injections numbered in order and joined in sequence; the ringed marker is the best injection. Random search scatters over the whole box; BO homes in on the dark band after its eight random starts; the LLM search begins with a broad scouting gradient and then moves one grid step at a time along the ridge.

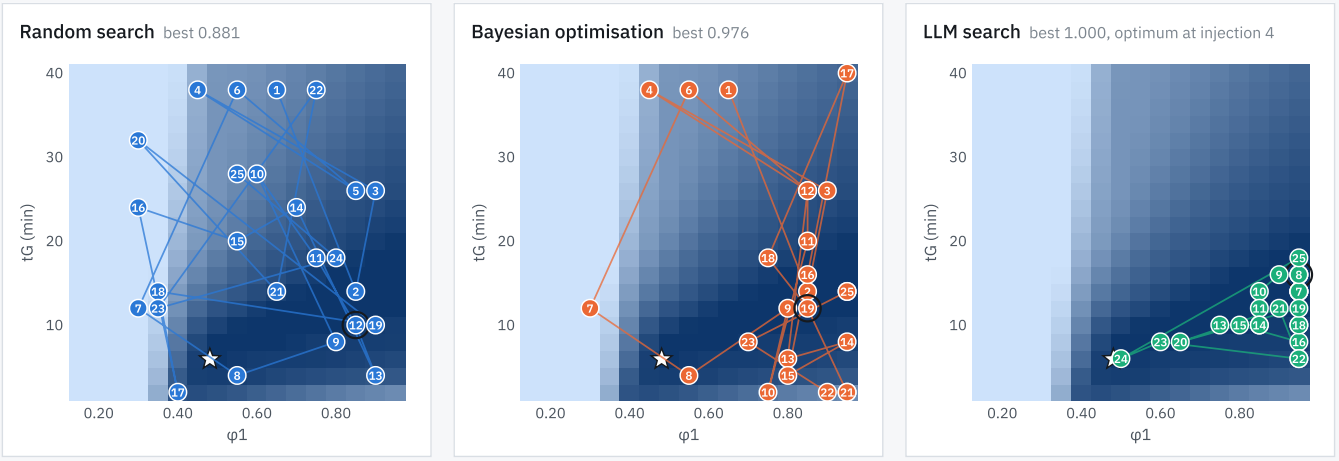
easy



medium



hard

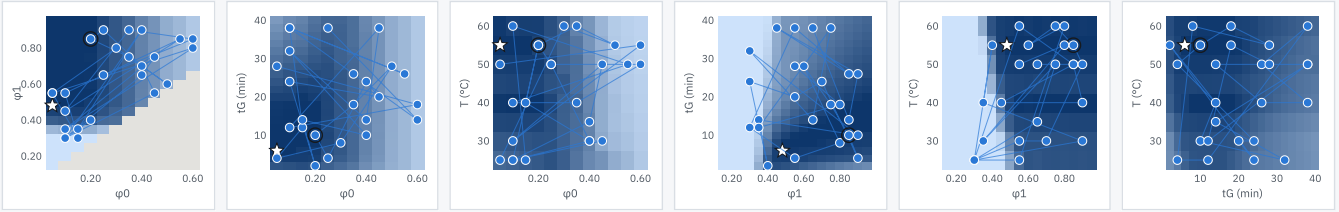


Random search — Bayesian optimisation — LLM search ★ stored optimum

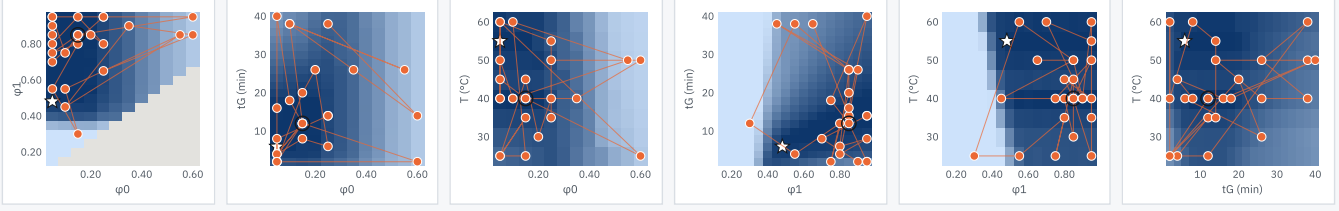
The hard set in all six projections

The same three seed-0 campaigns on the hard set, drawn on every pairwise panel of the landscape. The T column is where the methods differ most: random search and BO's first eight injections spread over every temperature, BO's later injections cluster at 35 to 45 °C around its best point, and the LLM search moves to 55 °C at injection 7 and stays there for all but three of the remaining injections.

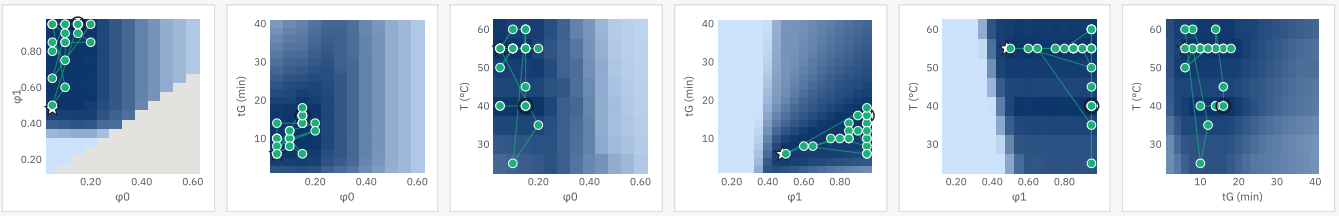
Random search · hard · seed 0 · best 0.881



Bayesian optimisation · hard · seed 0 · best 0.976



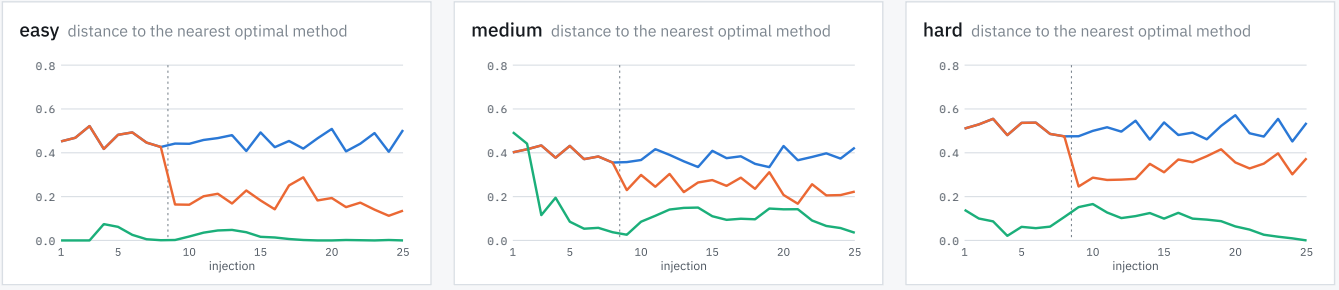
LLM search · hard · seed 0 · best 1.000



Navigation statistics over 50 campaigns

How far each search moves between consecutive injections (Euclidean distance in the unit cube, where the box diagonal is 2.0), how much of its budget it spends at or near the optimum, how often it runs a truncated method, and its mean injection score before and after injection 8, which is where BO's surrogate takes over.

set	method	mean step, all	mean step, after 8	injections at optimum	injections ≥ 0.9	truncated	mean CRF, inj 1–8	mean CRF, inj 9–25
easy	Random search	0.82	0.81	5 %	8 %	17 %	0.328	0.274
easy	Bayesian optimisation	0.75	0.72	27 %	43 %	6 %	0.328	0.835
easy	LLM search	0.19	0.15	79 %	97 %	0 %	0.984	0.995
medium	Random search	0.82	0.81	2 %	12 %	4 %	0.571	0.576
medium	Bayesian optimisation	0.80	0.79	13 %	29 %	2 %	0.571	0.773
medium	LLM search	0.29	0.31	31 %	76 %	0 %	0.895	0.950
hard	Random search	0.82	0.81	0 %	5 %	14 %	0.280	0.259
hard	Bayesian optimisation	0.75	0.71	2 %	24 %	7 %	0.280	0.664
hard	LLM search	0.29	0.34	50 %	93 %	0 %	0.963	0.958



Mean distance from injection n to the nearest grid method with CRF ≥ 0.99, in unit-cube units, averaged over 50 campaigns. Zero means the injection itself was optimal. Random search stays at the distance of a random draw; BO's distance falls once the surrogate takes over; the LLM search is near the ridge from its first injection and walks along it.

The LLM search, injection by injection

Every proposal the model made on seed 0, with the rationale it returned alongside the JSON. The hard set is expanded; the others fold.

▼ hard · seed 0 · optimum at injection 4 · 25 proposals, 0 re-asked for a repeat

#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
1	0.05	0.95	10	40	0.918	0.918	Start with a broad scouting gradient to observe all 10 compounds, locate their elution window, and identify critical pairs. A moderate temperature leaves room to explore temperature-dependent selectivity in both directions.
2	0.05	0.95	14	40	0.978	0.978	Change only gradient duration to measure how the two critical pairs respond to a shallower slope. The current last peak at 7.16 minutes leaves room to improve resolution while likely keeping all peaks within 10 minutes.

#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
3	0.15	0.95	14	40	0.980	0.980	The critical pair improved substantially with a shallower gradient. Raising phi0 narrows the gradient window at the same duration, further reducing the slope while trimming early low-organic time to help keep the last peak below 10 minutes.
4	0.15	0.95	16	40	**1.000**	1.000	Lengthen the best gradient by one grid step to push the remaining critical pair toward R ≥ 1.5. The last peak is expected near 9.6 minutes, preserving the zero time penalty.
5	0.15	0.95	16	45	0.920	1.000	Injection 4 reached the maximum CRF, so retain it as the benchmark. Raise temperature one step to seek a shorter fully resolved separation while measuring the critical pair's temperature sensitivity.
6	0.15	0.95	14	60	0.925	1.000	Test whether higher temperature carries the critical pair through its apparent selectivity crossover and restores separation. The shorter gradient also tests a faster alternative, while injection 4 remains the maximum-score fallback.
7	0.15	0.95	14	55	0.970	1.000	Step back from 60 °C to seek a temperature between the two observed selectivity bottlenecks. This may resolve the new close pair while preserving the improved separation of the original critical pair and a last peak below 8 minutes.
8	0.15	0.95	16	55	**0.994**	1.000	Lengthen the 55 °C gradient one step to improve the remaining close pair, while checking that the reordered pair retains adequate resolution. The last peak should remain comfortably below 10 minutes, potentially matching the maximum score with faster elution than injection 4.
9	0.15	0.90	16	55	**1.000**	1.000	Slightly narrow the gradient window to gain the small resolution increase needed for the R = 1.41 pair. The last peak elutes near 0.54 organic fraction, so this endpoint remains ample and should keep its retention below 10 minutes.
10	0.20	0.85	14	55	0.910	1.000	Shift the gradient window upward at its start while keeping nearly the same slope as injection 9, aiming to remove low-organic time without sacrificing critical-pair resolution. Check early-peak retention carefully because the first peak was already at 0.80 minutes.
11	0.10	0.85	12	55	**1.000**	1.000	The higher starting organic fraction damaged selectivity, so test a lower start to strengthen early retention and potentially separate the new critical pair. Combine this with a steeper, shorter gradient to seek full resolution faster than injection 9.
12	0.10	0.85	10	55	0.972	1.000	Injection 11 is fully resolved with the last peak at 8.00 minutes. Shorten its gradient by one grid step to test the speed limit while holding the successful starting composition and temperature fixed.
13	0.10	0.75	10	55	**1.000**	1.000	Use a slope between injections 11 and 12 by lowering the endpoint while retaining the 10-minute gradient. This should recover full resolution of the slope-sensitive pair with a last peak slightly earlier than injection 11.
14	0.05	0.85	10	55	**0.990**	1.000	Test whether a lower starting organic fraction improves the slope-sensitive pair enough to permit a steeper gradient. This could shorten elution relative to injection 13, while checking that the other critical pair remains resolved.
15	0.05	0.80	10	55	**1.000**	1.000	Reduce the slope slightly from injection 14 to lift its R = 1.37 pair above 1.5. The other critical pair has responded weakly to slope, so this may retain full resolution with the last peak earlier than injection 13.
16	0.10	0.95	8	60	0.941	1.000	Retain injection 15 as the fastest fully resolved benchmark and explore a faster high-temperature regime. At 60 °C the middle pair has crossed elution order; a steeper gradient may separate it on the opposite side of that crossover.
17	0.15	0.95	6	60	0.914	1.000	Test a steeper gradient and higher starting organic fraction to widen the crossed middle pair at 60 °C. Monitor the neighboring pairs for compression; injection 15 remains the fastest confirmed fully resolved method.
18	0.10	0.95	10	25	0.690	1.000	Explore low-temperature selectivity, since the fast 60 °C gradients trade one unresolved pair for another. A broad 10-minute gradient should keep retention manageable while testing whether reordered pairs offer a faster fully resolved alternative.
19	0.20	0.95	12	35	0.915	1.000	Test intermediate low-temperature selectivity with a higher starting organic fraction to limit retention time. This explores whether 35 °C separates the central cluster without the severe crowding seen at 25 °C, while retaining the confirmed maximum-score methods as fallbacks.
20	0.05	0.65	8	55	**1.000**	1.000	Trim the unused high-organic portion of injection 15 while preserving its successful gradient slope exactly. Its last peak eluted near 0.57 organic fraction, so a 0.65 endpoint should retain full resolution with a shorter programmed gradient.
21	0.10	0.90	12	55	**0.998**	1.000	Test a small slope increase over injection 13, from 0.065 to 0.0667 organic fraction per minute, to refine the boundary of full resolution. This may shorten the separation while preserving the maximum CRF; injection 20 remains the fastest confirmed fallback.
22	0.05	0.95	6	50	0.883	1.000	Test the remaining intermediate-temperature regime with a steep gradient, which may separate the original critical pair after its selectivity crossover. The low

#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
							starting organic fraction protects early retention; injection 20 remains the fastest confirmed maximum-score method.
23	0.10	0.60	8	55	**1.000**	1.000	Trim injection 11 to an 8-minute gradient while preserving its fully resolving slope and starting composition. Its last peak should remain inside the gradient, offering a compact maximum-score method with slightly more resolution margin than injection 20.
24	0.05	0.50	6	55	**1.000**	1.000	Preserve injection 20's resolving slope but end the gradient after the first nine peaks, testing a shorter programmed gradient. The isolated last compound will elute during the hold; check whether it still arrives before 10 minutes to preserve the maximum CRF.
25	0.15	0.95	18	55	**1.000**	1.000	Use the final injection to seek a larger resolution margin at the maximum CRF: extrapolating injections 8–9 suggests both critical pairs near R = 1.7 and the last peak below 10 minutes. Injection 20 remains the fastest confirmed fully resolved method.

▼ medium · seed 0 · optimum at injection 5 · 25 proposals, 0 re-asked for a repeat

#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
1	0.05	0.95	12	40	0.659	0.659	Start with a broad gradient at an intermediate temperature to locate the elution window and critical pairs while minimizing the risk of missing retained compounds. This provides a baseline for targeted changes in gradient slope and temperature.
2	0.05	0.95	20	40	0.874	0.874	Change only gradient duration to measure which crowded pairs respond to a shallower slope. The initial run leaves enough time margin to spread the central cluster and late pair without a substantial time penalty.
3	0.05	0.95	20	60	0.962	0.962	The shallower gradient resolved most of the crowding but left one nearly coeluting pair. Raising temperature while holding the gradient fixed tests selectivity for that pair and should shorten retention.
4	0.05	0.95	22	60	0.908	0.962	Slightly flatten the gradient at 60 C to improve the remaining close pair while retaining the favorable temperature selectivity. The last peak at 8.56 minutes leaves room for this adjustment before the time penalty begins.
5	0.05	0.95	18	60	**0.996**	0.996	The remaining close pair worsened with a shallower gradient, so reverse direction and steepen slightly relative to the best run. This should improve that pair and shorten the run, while testing whether the earlier pairs retain adequate resolution.
6	0.05	0.95	18	55	0.860	0.996	The best run is almost fully resolved, so test a small temperature adjustment while preserving its gradient. This probes whether modest selectivity changes can lift the R = 1.46 pair above threshold without compromising the other pairs.
7	0.10	0.95	18	60	0.985	0.996	Return to 60 C and raise the starting organic fraction one step, which slightly shallows the gradient while shifting its window. This tests a finer selectivity adjustment between the competing early and late critical pairs than changing duration by two minutes.
8	0.10	0.90	18	60	**0.999**	0.999	Lower the endpoint one step to gently flatten injection 7's gradient, aiming to resolve its two early close pairs. Its later critical pair has some resolution margin, making this a promising balance between the competing slope responses.
9	0.10	0.55	10	60	**1.000**	1.000	Use a slope of 0.045 organic fraction per minute, between injections 7 and 8, to recover the late pair's resolution while preserving the early separations. The narrower window still comfortably covers all observed elution fractions and permits this finer slope adjustment on the instrument grid.
10	0.15	0.55	10	60	0.960	1.000	Injection 9 has reached the maximum CRF and remains the benchmark. Raise the starting organic fraction while keeping the endpoint and duration fixed to test a shorter elution window, with the shallower slope helping offset compression of the early pairs.
11	0.10	0.65	12	60	**0.999**	1.000	Return to the fully resolved starting fraction and test a slightly steeper slope than injection 9 to shorten retention. This small grid-compatible adjustment probes the resolution limit while retaining injection 9 as the maximum-score benchmark.
12	0.10	0.55	10	25	0.969	1.000	Injection 9 remains optimal in CRF, and further steepening has reached a resolution limit. Test the low-temperature selectivity regime with its gradient unchanged to look for an alternative separation pattern that could support subsequent acceleration.
13	0.10	0.55	10	30	0.943	1.000	The low-temperature run has only one unresolved pair, so raise temperature one step with the gradient unchanged to test its selectivity response. If full resolution is recovered, this alternative temperature regime can then be tested with a narrower, faster elution window.
14	0.20	0.65	10	25	0.892	1.000	Shift the low-temperature gradient window upward while preserving its slope to shorten retention and test starting-fraction selectivity for the remaining close pair.

#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
							Injection 9 remains the maximum-CRF benchmark.
15	0.20	0.85	10	25	0.920	1.000	Test a steeper gradient in the shifted low-temperature window to move the nearly coeluting pair through its apparent selectivity crossover and shorten retention. This explores a faster alternative while preserving injection 9 as the established maximum-score method.
16	0.30	0.90	10	25	**1.000**	1.000	Raise the starting fraction to test whether the central pair separates further after its observed order reversal. Slightly reduce the slope relative to injection 15 to help the late pair, while the higher starting fraction should shorten retention.
17	0.35	0.95	10	25	0.973	1.000	Injection 16 achieves maximum CRF with much shorter retention. Shift its gradient window upward one step at unchanged slope to test further acceleration while monitoring the central cluster and the marginally resolved late pair.
18	0.35	0.85	10	25	**1.000**	1.000	Flatten injection 17's gradient to recover its two under-resolved pairs while retaining the faster starting fraction. This may achieve full resolution with a last peak earlier than injection 16's 4.56 minutes.
19	0.40	0.70	10	25	0.941	1.000	Test a higher starting fraction with a substantially shallower slope to offset the central-pair compression seen when shifting the window upward. Injection 18 remains the fastest fully resolved benchmark; this tests whether another window shift can shorten retention without losing resolution.
20	0.35	0.85	10	30	0.930	1.000	Return to injection 18's fully resolved window and raise temperature one step to test further acceleration through temperature rather than starting fraction. Monitor the central pair closely, since its resolution margin is modest.
21	0.35	0.85	10	45	0.762	1.000	The central pair converges as temperature rises from 25 to 30 C; test beyond the likely crossover to see whether it separates again in reversed order. This explores a faster selectivity regime while retaining injection 18 as the fastest fully resolved method.
22	0.35	0.90	10	25	**0.991**	1.000	Return to the fastest fully resolved temperature and starting fraction, then test a slope halfway between injections 18 and 17. This brackets the steepest usable gradient for this window while keeping injection 18 as the fully resolved fallback.
23	0.40	0.60	10	25	0.973	1.000	Test whether a shallower slope can rescue the single unresolved pair at the higher starting fraction used in injection 19. Injection 18 remains the fastest fully resolved benchmark; this run checks the remaining nearby window before final selection.
24	0.40	0.50	10	25	**1.000**	1.000	The sole unresolved pair at phi0 = 0.40 improves as the slope decreases, so flatten it another step to test for full resolution. This may provide a higher-resolution alternative, while injection 18 remains the fastest fully resolved benchmark.
25	0.35	0.55	4	25	**1.000**	1.000	Trim injection 18's gradient to just beyond its observed elution window while preserving its starting fraction, slope, and temperature. All peaks should still elute during the gradient, retaining maximum CRF and the 4.14-minute last peak while shortening the programmed gradient from 10 to 4 minutes.

▼ easy · seed 0 · optimum at injection 1 · 25 proposals, 0 re-asked for a repeat

#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
1	0.05	0.95	10	40	**1.000**	1.000	Start with a broad gradient to locate the six-peak elution window and identify critical pairs while minimizing the risk of missing strongly retained compounds. The moderate temperature leaves room to adjust selectivity in either direction.
2	0.05	0.95	6	40	**1.000**	1.000	Injection 1 already achieved the maximum CRF, with substantial resolution margin. Shorten the gradient while holding the solvent window and temperature fixed to seek a faster fully resolved method.
3	0.05	0.95	4	40	**1.000**	1.000	The six-minute gradient retains full resolution, with the closest pair at R = 2.46. Test the next shorter gradient to reduce runtime while preserving the maximum CRF.
4	0.05	0.95	2	40	0.909	1.000	Test the minimum gradient duration to establish the speed–resolution boundary. The first pair may fall below R = 1.5; the four-minute method remains a fully resolved fallback with maximum CRF.
5	0.05	0.95	2	60	0.764	1.000	Test whether higher temperature improves selectivity for the unresolved first pair at the minimum gradient duration. This isolates the temperature effect and may recover full resolution with an even earlier last peak.
6	0.05	0.95	2	25	0.964	1.000	Higher temperature worsened the critical separations, so test the lowest available temperature at the same gradient slope. This should favor resolution and establish whether temperature alone can rescue the two-minute gradient.

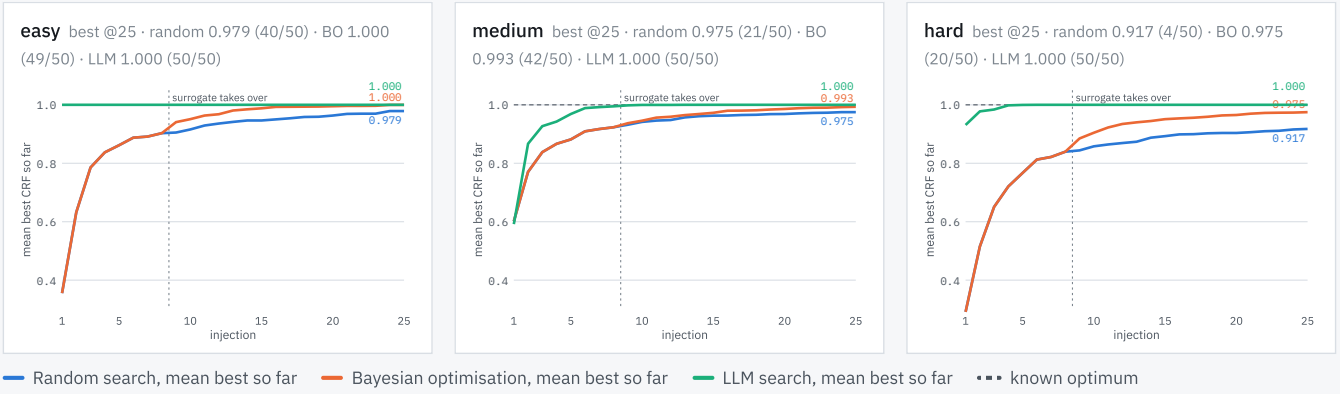
#	φ0	φ1	tG	T	CRF	best	rationale (model's own words)
7	0.05	0.75	2	25	**1.000**	1.000	Reduce gradient steepness at the temperature that best separates the first pair, aiming to lift its resolution above 1.5. The last compound will likely elute during the final hold, allowing a short gradient without sacrificing early-pair resolution.
8	0.15	0.85	2	25	**1.000**	1.000	Shift the successful two-minute gradient window upward while preserving its slope and temperature. This tests whether stronger final solvent can accelerate the last peak while retaining full resolution of the first pair.
9	0.25	0.95	2	25	**1.000**	1.000	Shift the gradient window upward another 0.10 at unchanged slope, which preserved full resolution in the previous shift. The stronger endpoint should shorten the final hold, while this run tests whether the first pair remains sufficiently retained and resolved.
10	0.30	0.95	2	25	**1.000**	1.000	Raise the starting organic fraction one step to shorten retention while slightly flattening the gradient. Check whether the earliest pair retains full resolution as it approaches the weakly retained region.
11	0.30	0.95	2	30	**1.000**	1.000	Raise temperature by one grid step to shorten the run, using the first pair's resolution margin of 1.77. Keep the gradient unchanged to measure whether the faster method still achieves maximum CRF.
12	0.30	0.95	2	35	0.981	1.000	Test the next temperature step to locate the fastest fully resolved temperature for this gradient. The first pair is near its resolution limit, so injection 11 remains the maximum-CRF fallback if this step loses separation.
13	0.35	0.95	2	25	0.984	1.000	Return to the temperature that favors the first pair and test one higher starting-organic step. This probes whether a narrower gradient can shorten the run further without compressing the early peaks below R = 1.5.
14	0.30	0.90	2	30	**1.000**	1.000	The fastest fully resolved method so far is injection 11, but its first pair has little resolution margin. Lower the endpoint one step to flatten the gradient and seek a more robust maximum-CRF method with only a modest increase in final-peak retention.
15	0.25	0.85	2	35	0.989	1.000	Test whether a lower starting fraction and shallower gradient restore first-pair resolution at 35 C. This explores an alternative maximum-CRF window while keeping the earliest peaks away from the near-unretained region.
16	0.25	0.80	2	35	**1.000**	1.000	Flatten the previous gradient by lowering its endpoint one step, aiming to raise the first pair from R = 1.42 to at least 1.5. This should keep the last peak comfortably below ten minutes and may establish a fully resolved two-minute gradient at 35 C.
17	0.25	0.90	2	25	**1.000**	1.000	Injection 16 is borderline despite its rounded score, so return to 25 C and test a window between two successful methods. This targets maximum CRF with more first-pair resolution margin while keeping the final peak near three minutes.
18	0.30	0.90	2	25	**1.000**	1.000	Combine the favorable starting fraction from injection 10 with the shallower slope tested in injection 14. This should preserve maximum CRF and improve the first-pair resolution margin while keeping the last peak below three minutes.
19	0.25	0.90	2	30	**1.000**	1.000	Raise temperature one step from injection 17 to shorten final-peak retention while testing whether its first-pair resolution margin is sufficient. This also checks a lower-starting-organic alternative to injection 14, with better retention of the earliest peaks.
20	0.25	0.85	2	25	**1.000**	1.000	Test a shallower neighboring method at the temperature most favorable to the critical pair, seeking additional resolution margin with all peaks still near three minutes. Injection 11 remains the fastest confirmed fully resolved method; the maximum CRF has already been reached.
21	0.25	0.85	2	30	**1.000**	1.000	Test the intermediate temperature between injection 20's full separation and injection 15's unresolved first pair. This should shorten the last-peak time relative to injection 20 while retaining maximum CRF and more resolution margin than injection 19.
22	0.30	0.85	2	30	**1.000**	1.000	Raise the starting fraction one step from injection 21 to reduce retention while retaining a shallow gradient. Neighboring runs suggest this may preserve full resolution and slightly improve the critical first-pair margin.
23	0.30	0.85	2	25	**1.000**	1.000	Lower temperature one step from injection 22 to increase the critical first-pair resolution margin while keeping the run short and targeting maximum CRF. Injection 11 remains the fastest confirmed fully resolved method.
24	0.20	0.95	2	25	**1.000**	1.000	Test a neighboring high-endpoint method that should keep the final peak close to the fastest runs while retaining the first peak longer. The first-pair resolution is predicted to be near 1.5, so this checks the lower-starting-organic boundary of the short, maximum-CRF region.
25	0.20	0.90	2	25	**1.000**	1.000	Lower the endpoint one step from injection 24 to improve its marginal first-pair resolution while keeping the last peak near three minutes. The maximum CRF is already secured, and injection 11 remains the fastest confirmed fully resolved method.

8. Results: fifty campaigns per method, cold start

Each method was run 50 times per set with 25 injections per campaign. For random search and BO the campaign number is the seed of the row permutation; for the LLM search it only labels the repeat, since the model's proposals are nondeterministic and every campaign starts from a blank page. "Best @25" is the best CRF a campaign had after its 25th injection, averaged over the 50 repeats. "Found optimum" counts the campaigns whose best was within 1 % of the optimum, which on these sets means CRF ≥ 0.99. "Median injections to optimum" is taken over the campaigns that found it.

> On the hard set the LLM search reached the optimum in **50 of 50** campaigns (mean best @25 1.000), against 20 of 50 for BO (0.975) and 4 of 50 for random search (0.917). On medium: 50/50, 42/50 and 21/50; on easy: 50/50, 49/50 and 40/50. Median injections to the optimum on hard, among campaigns that found it: LLM 4, BO 14, random 7.

Set	Method	Mean best CRF	Found optimum	Median injections to optimum
easy	Random search	**0.979**	**40/50** (80 %)	10
easy	Bayesian optimisation	**1.000**	**49/50** (98 %)	9
easy	LLM search	**1.000**	**50/50** (100 %)	1
medium	Random search	**0.975**	**21/50** (42 %)	13
medium	Bayesian optimisation	**0.993**	**42/50** (84 %)	15
medium	LLM search	**1.000**	**50/50** (100 %)	6
hard	Random search	**0.917**	**4/50** (8 %)	7
hard	Bayesian optimisation	**0.975**	**20/50** (40 %)	14
hard	LLM search	**1.000**	**50/50** (100 %)	4



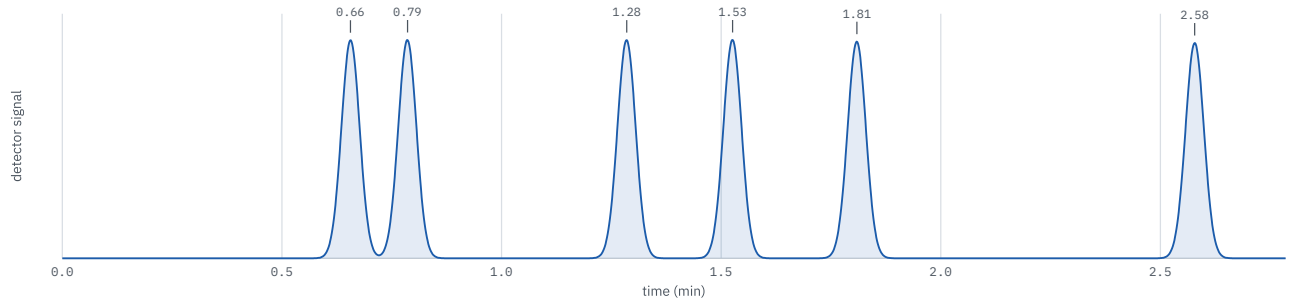
Mean best CRF so far against injection number, 50 campaigns per method and set. Random search and BO coincide for the first eight injections, which are the same random rows, and separate when the surrogate takes over. The LLM curve starts from its own first scouting gradient.

set	method	mean best @5 (share at optimum)	mean best @10 (share at optimum)	mean best @15 (share at optimum)	mean best @20 (share at optimum)	mean best @25 (share at optimum)
easy	Random search	0.862 (28 %)	0.916 (46 %)	0.947 (58 %)	0.963 (68 %)	0.979 (80 %)
easy	Bayesian optimisation	0.862 (28 %)	0.951 (62 %)	0.988 (86 %)	0.995 (96 %)	1.000 (98 %)
easy	LLM search	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)
medium	Random search	0.881 (2 %)	0.942 (16 %)	0.963 (30 %)	0.968 (34 %)	0.975 (42 %)
medium	Bayesian optimisation	0.881 (2 %)	0.946 (20 %)	0.973 (48 %)	0.986 (70 %)	0.993 (84 %)
medium	LLM search	0.968 (30 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)
hard	Random search	0.767 (2 %)	0.858 (6 %)	0.893 (6 %)	0.904 (8 %)	0.917 (8 %)
hard	Bayesian optimisation	0.767 (2 %)	0.905 (12 %)	0.951 (26 %)	0.965 (30 %)	0.975 (40 %)
hard	LLM search	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)

The best LLM method, as the detector would show it

For each set, the LLM search's best injection (CRF 1.0; among ties, the earliest last peak, which is the criterion the model itself used). The trace is the simulator's detector view: a sum of Gaussians, one per analyte with unit area, sampled at 10 Hz, noise-free. Every peak is baseline-resolved (all adjacent R ≥ 1.5) and the last one elutes inside 10 minutes, so no time penalty applies. Hover a peak apex for its retention time and width.

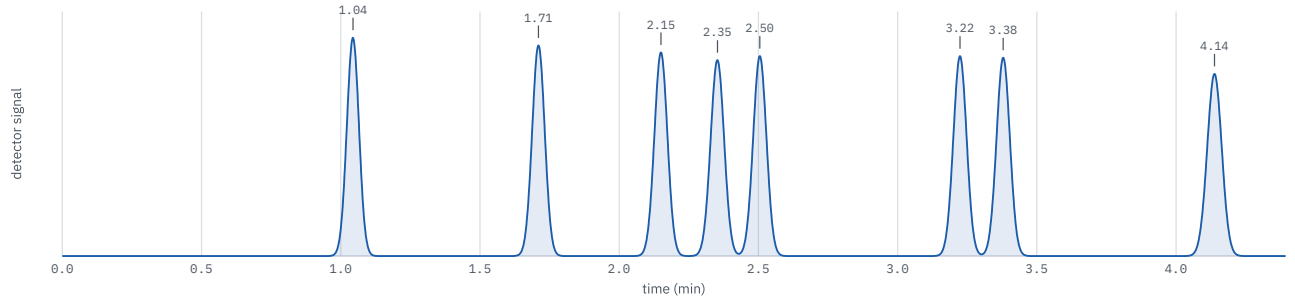
easy · injection 11 φ0 0.30 → φ1 0.95 over 2 min at 30 °C · CRF 1.000 · last peak 2.58 min · min R 1.58



row	1	2	3	4	5	6
t_R (min)	0.66	0.79	1.28	1.53	1.81	2.58
σ (min)	0.0205	0.0205	0.0205	0.0205	0.0207	0.0208
R to next	**1.58**	6.08	2.94	3.44	9.28	–

Model's rationale for this injection: “Raise temperature by one grid step to shorten the run, using the first pair’s resolution margin of 1.77. Keep the gradient unchanged to measure whether the faster method still achieves maximum CRF.”

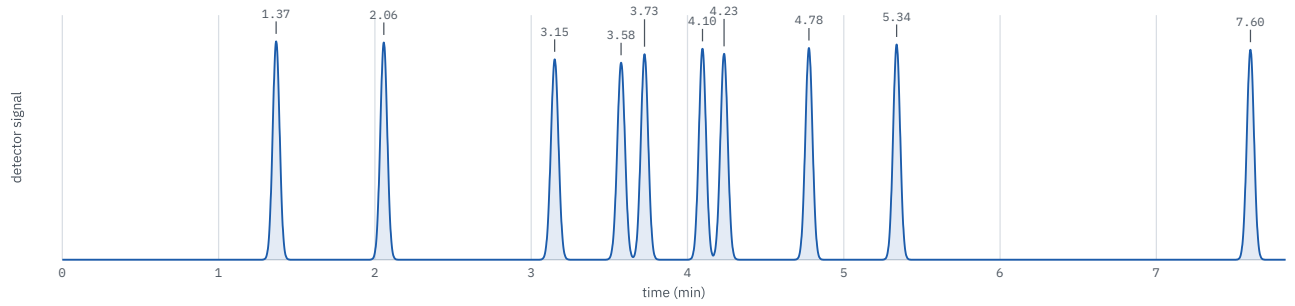
medium · injection 25 φ0 0.35 → φ1 0.55 over 4 min at 25 °C · CRF 1.000 · last peak 4.14 min · min R 1.61



row	1	2	3	4	5	6	7	8
t_R (min)	1.04	1.71	2.15	2.35	2.50	3.22	3.38	4.14
σ (min)	0.0214	0.0222	0.0229	0.0238	0.0234	0.0234	0.0235	0.0257
R to next	7.64	4.87	2.17	1.61	7.69	1.65	7.71	–

Model's rationale for this injection: “Trim injection 18’s gradient to just beyond its observed elution window while preserving its starting fraction, slope, and temperature. All peaks should still elute during the gradient, retaining maximum CRF and the 4.14-minute last peak while shortening the programmed gradient from 10 to 4 minutes.”

hard · injection 15 φ0 0.05 → φ1 0.80 over 10 min at 55 °C · CRF 1.000 · last peak 7.60 min · min R 1.52



row	1	2	3	4	5	6	7	8	9	10
t_R (min)	1.37	2.06	3.15	3.58	3.73	4.10	4.23	4.78	5.34	7.60
σ (min)	0.0217	0.0218	0.0236	0.0241	0.0231	0.0225	0.0230	0.0224	0.0220	0.0226
R to next	7.91	12.03	4.46	**1.59**	4.07	**1.52**	5.98	6.32	25.39	–

Model's rationale for this injection: “Reduce the slope slightly from injection 14 to lift its R = 1.37 pair above 1.5. The other critical pair has responded weakly to slope, so this may retain full resolution with the last peak earlier than injection 13.”

9. Real compounds: two sets from the HPLC Simulator library

The three sets above are random draws, so a search that does well on them has only been shown to do well on the generator's idea of a mixture. This section repeats the benchmark on measured compounds. The HPLC Simulator (Boswell, Stoll, Carr et al., *J. Chem. Educ.* 2013, 90, 198) ships a library of 43 compounds measured on two reversed-phase columns, each with a retention surface in both organic fraction and temperature: $\log_{10} k = (w + m{\cdot}T) + (v + n{\cdot}T){\cdot}\varphi$, with T in °C and separate coefficients for acetonitrile and methanol. The library is redistributed in the project under its CC BY-NC-SA 3.0 licence, and everything derived from it lives in its own data root, `hplcsim/`, laid out like the project root.

Fitting the library to the simulator. The simulator's model, $\log_{10} k = \log k_w + \Delta H\,(1/T_{ref} - 1/T) - S{\cdot}\varphi$, has a slope S that does not change with temperature; the library's slope does (its n term). So each compound is fitted by weighted linear least squares over the whole method box (φ 0.05 to 0.95, T 25 to 60 °C), with full weight where the true retention factor lies between 0.5 and 50, which is the band in which a gradient actually moves the compound, and a tenth of that weight elsewhere. The three fitted numbers go into the analyte YAML exactly like a generated analyte; the residuals go into a sidecar file. Both sets then run through the same optimum finder, table builder and optimisers with no code change.

set	source phase / modifier	n	log k_w	S	ΔH (K)	median fit RMSE	worst fits (RMSE in log k, inside the window)
`hplcsim_beh`	Waters Acquity BEH C18 / acetonitrile	21	0.57 to 3.35	2.8 to 6.1	−816 to +307	0.007	caffeine 0.119 · 2-acetylfuran 0.048 · acetanilide 0.032
`hplcsim_sb`	Agilent Zorbax SB-C18 / acetonitrile	22	0.65 to 3.78	1.7 to 4.4	−670 to −309	0.019	p-nitrobenzyl chloride 0.026 · naphthalene 0.024 · bromobenzene 0.023

For comparison the generated sets draw log k_w from 1.8 to 4.0, S from 4 to 8 and ΔH from −3000 to −1000. The real compounds have shallower solvent slopes and a weaker temperature dependence; caffeine and 2-acetylfuran are the two whose slope drifts so much with temperature that the fit returns a near-zero or positive ΔH, and they are the two weakly retained compounds that elute first in every method.

What the two sets ask of a search

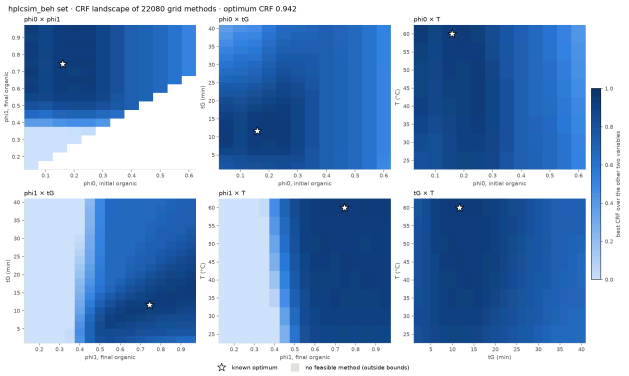
Both are larger mixtures than any generated set, and neither reaches CRF 1.0 within the 42 min window: the stored optima are 0.942 and 0.945, so the ceiling is the chemistry, not a hidden corner of the grid. The region within 1 % of the optimum is also an order of magnitude smaller than the hard set's: 11 and 15 grid rows out of 22 080, against 82 on hard.

set	rows	truncated rows	median CRF	best grid CRF	stored optimum	optimum method	rows within 1 % of optimum	rows ≥ 0.9
`hplcsim_beh`	22 080	11.0 %	0.565	0.938	0.942	φ 0.16 → 0.75 over 11.6 min at 60 °C	11 (0.05 %)	263
`hplcsim_sb`	22 080	16.2 %	0.536	0.944	0.945	φ 0.28 → 0.84 over 12.3 min at 25 °C	15 (0.07 %)	255
hard, for scale	22 080	12.7 %	–	1.000	1.000	φ 0.05 → 0.48 over 5.9 min at 55 °C	82 (0.37 %)	–

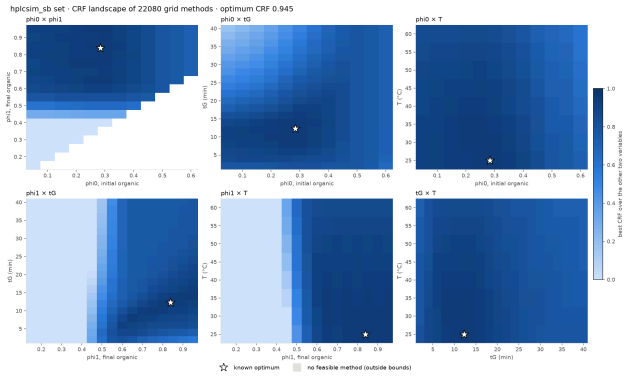
CRF landscape

The same six pairwise max-projections as in section 4, drawn by the project's own plotting code: each cell is the best CRF over the two variables not on the axes, the star is the stored optimum, and the white triangle holds no method because φ_1 must exceed φ_0 by 0.10.

hplcsim_beh 21 compounds · BEH C18



hplcsim_sb 22 compounds · SB-C18

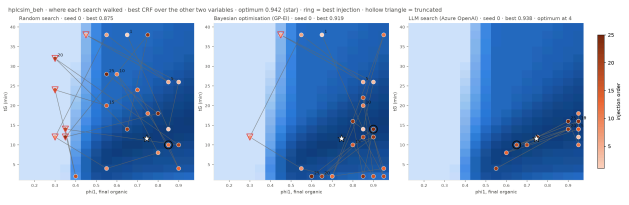


Three things are common to both sets and differ from the generated ones. First, the final composition is a cliff: below $\varphi_1 \approx 0.5$ the best CRF collapses from about 0.9 to below zero, because the most retained compounds (octanophenone and anthracene on BEH, n-butylbenzene on SB) no longer leave the column within 42 min. Second, φ_0 and t_G form a ridge rather than a peak: a low start with a short gradient scores the same as a higher start with a longer one, and the time penalty beyond 10 min tilts the ridge toward about 12 min. Third, temperature matters on one set and hardly at all on the other: on BEH the best CRF rises monotonically to the 60 °C bound, on SB it is flat and drifts from 0.945 at 25 °C down to about 0.85 at 60 °C. There is no dark cell at 1.0 anywhere, which is what the ceiling of 0.94 looks like.

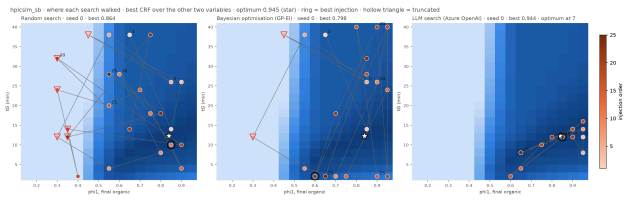
Where each search walked

The seed-0 campaign of each method drawn on the $\varphi_1 \times t_G$ projection of the landscape, as in section 7 for the generated sets. Markers are injections coloured by order, the ring is the best injection, hollow triangles are truncated runs. The full corner plots per method are in `hplcsim/results/hplcsim_*_trajectory_{random,bo,llm}.png`.

hplcsim_beh random · BO · LLM, seed 0



hplcsim_sb random · BO · LLM, seed 0



Random search draws from the whole box and pays for it with truncated runs at low φ_1. BO's eight random starts are the same rows; its surrogate then pulls it to high φ_1, but on the SB set it spends most of the remaining budget along the 2 min edge, where the time term is zero but the near pairs do not resolve, and ends at 0.798. The LLM search never leaves the ridge: on BEH it walks from the 0.95 scouting window down to φ_1 0.65 at 10 min, on SB it circles the optimum within two grid steps from injection 5 on.

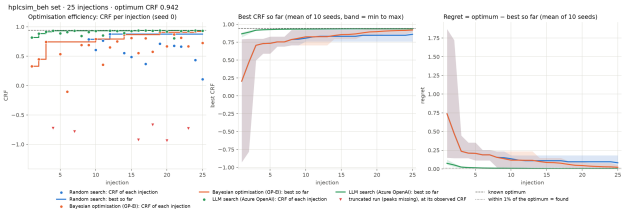
How the searches perform

Budget 25 injections as before, and ten campaigns per method and set: random search and BO with seeds 0 to 9, the LLM search ten times from a blank page (the seed only labels the repeat), with the same system prompt as in section 5 and no change for the new sets, which also makes this the memorisation check the design called for: nothing about these compounds was ever in a prompt. "Found" again means a best CRF within 1 % of the stored optimum.

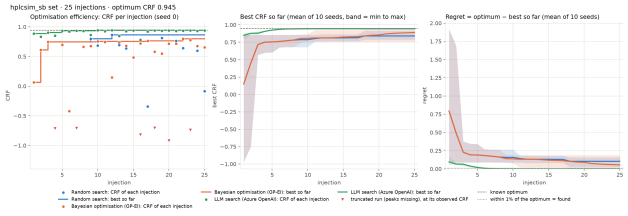
set	method	campaigns	mean best @25	worst campaign	best campaign	found optimum	injections to find it	mean regret @25
hplcsim_beh optimum 0.942	random	10	0.860	0.761	0.931	0 of 10	–	0.082
	BO	10	0.920	0.890	0.938	1 of 10	19	0.021
	LLM	10	0.938	0.938	0.938	**10 of 10**	**6.5 (3 to 13)**	0.004
hplcsim_sb optimum 0.945	random	10	0.841	0.787	0.915	0 of 10	–	0.104
	BO	10	0.891	0.757	0.944	1 of 10	23	0.054
	LLM	10	0.944	0.944	0.944	**10 of 10**	**6 (5 to 7)**	0.001

mean best CRF so far, after injection	1	5	10	15	20	25
hplcsim_beh · random (10)	0.205	0.734	0.806	0.832	0.847	0.860
hplcsim_beh · BO (10)	0.205	0.734	0.825	0.856	0.897	0.920
hplcsim_beh · LLM (10)	0.867	0.927	0.934	0.937	0.938	0.938
hplcsim_sb · random (10)	0.150	0.755	0.787	0.813	0.841	0.841
hplcsim_sb · BO (10)	0.150	0.755	0.807	0.825	0.855	0.891
hplcsim_sb · LLM (10)	0.850	0.925	0.944	0.944	0.944	0.944

hplcsim_beh seed 0 per injection · mean of 10 seeds



hplcsim_sb seed 0 per injection · mean of 10 seeds



Left of each figure: seed 0, CRF of every injection with the best-so-far step line; red triangles are truncated runs (peaks still on the column at 42 min). Middle and right: mean best CRF and mean regret over the 10 campaigns of each method, with the min to max band; the LLM band collapses to a line after injection 13 because all ten campaigns end on the same best grid method. Random search lost peaks in 14 % (BEH) and 18 % (SB) of its injections; the LLM search ran no truncated method in 500.

On both sets the LLM search opened with a full scouting gradient (φ 0.05 → 0.95), read the elution window and the critical pairs from the first table, and then moved along one axis at a time. On BEH it tried 60 °C at injection 3, kept it, and spent the rest of the budget raising φ_0 and trimming φ_1. On SB it tested 60 °C and 25 °C in injections 2 and 3, found the cold end better, and converged on φ_0 0.25 to 0.30 with a 12 to 14 min gradient. BO reached the same region in one seed out of ten on each set, at injection 19 and 23; random search never did. Over the ten LLM campaigns every one reached the optimum region, at injection 3 to 13 on BEH (median 6.5) and 5 to 7 on SB (median 6), and all ten per set ended on the same best grid method: the spread is in how fast, not in where.

▼ LLM search on hplcsim_beh, seed 0, injection by injection

#	φ_0	φ_1	t_G	T	CRF	best	the model's rationale, as returned with the proposal
1	0.05	0.95	16	40	0.818	0.818	Use a broad scouting gradient to locate the full elution window while minimizing the risk of missing strongly retained compounds. An intermediate temperature leaves room to explore temperature-dependent selectivity in either direction.
2	0.05	0.95	12	40	0.886	0.886	Shorten only the gradient to test which critical pairs respond to slope while bringing the last peak closer to the 10-minute target. Keeping the solvent endpoints and temperature fixed makes this a clean comparison with the scouting run.
3	0.05	0.95	12	60	0.913	0.913	Test a substantial temperature increase at the same gradient to probe selectivity in the unresolved clusters, especially the late triplet. Reduced retention may also create room for a shallower gradient without exceeding 10 minutes.

#	φ_0	φ_1	t_G	T	CRF	best	the model's rationale, as returned with the proposal
4	0.05	0.95	14	60	0.933	**0.933**	Use the retention-time headroom at 60 C to test a shallower gradient, with the last peak expected near 10 minutes. This isolates the slope response of the remaining early pair and late cluster at the more promising temperature.
5	0.10	0.95	14	60	0.887	**0.933**	Raise the starting organic fraction one step to remove low-organic time while making the gradient slightly shallower. This may improve the remaining slope-sensitive pairs and bring the last peak below 10 minutes; the early peaks will be checked for compression.
6	0.20	0.95	14	60	0.911	**0.933**	Test whether a higher starting fraction moves the early pairs beyond their apparent selectivity crossings while improving late-pair resolution through a shallower slope. This is an exploratory step; monitor the first peaks for excessive compression.
7	0.20	0.95	16	60	0.934	**0.934**	Spend the available time headroom on a shallower gradient, targeting a last peak near 10 minutes. Holding the starting fraction and temperature fixed will reveal whether the unresolved early cluster improves along with the late pair.
8	0.20	0.95	16	25	0.846	**0.934**	Probe the opposite temperature limit to test whether selectivity can rescue the nearly slope-insensitive early pair. Keeping the gradient unchanged isolates temperature effects and reveals whether the late cluster separates on the other side of its order crossing.
9	0.25	0.95	18	60	0.912	**0.934**	Combine a higher starting fraction with a shallower gradient to improve the late pair while keeping the last peak near 10 minutes. This also tests whether the early cluster separates farther from its crossing, although compression of the first three peaks is a risk.
10	0.15	0.95	16	60	0.926	**0.934**	Test the intermediate starting fraction to balance early-cluster selectivity against late-pair resolution. This brackets the promising 0.20-start method while retaining a shallow gradient and a last peak expected close to 10 minutes.
11	0.20	0.95	10	25	0.855	**0.934**	Explore the steeper-gradient side of the late-cluster crossing at low temperature, where the early cluster showed better selectivity. This sacrifices some slope-dependent resolution but removes the time penalty and may separate the late peaks in a different order.
12	0.20	0.90	16	60	0.933	**0.934**	Refine the best method with a slightly shallower gradient by lowering the endpoint one grid step. This should improve the late critical pair and bring the middle pair above 1.5, with the last peak expected close to 10 minutes.
13	0.05	0.60	8	60	0.927	**0.934**	Test an earlier transition to the final hold, using a gradient slope close to the successful low-start methods. The hold may change late-pair selectivity independently of the early separation, though it risks a longer last-peak retention time.
14	0.15	0.95	14	40	0.894	**0.934**	Test intermediate temperature at the 0.15 starting fraction to shift the early cluster away from its high-temperature crossing. The shorter gradient should avoid a time penalty while assessing whether this selectivity region merits refinement.
15	0.20	0.95	16	55	0.929	**0.934**	Lower temperature one step from the best method to test whether improved early-cluster selectivity outweighs any loss in the late pair. The current retention-time margin should largely absorb the small increase in retention.
16	0.20	0.75	12	60	0.936	**0.936**	Choose a slope between injections 7 and 12 to approach 10 minutes without a time penalty. The endpoint remains above the observed elution window, so this should preserve gradient elution while improving the late critical pair over injection 7.
17	0.20	0.65	10	60	0.938	**0.938**	Refine the current best slope from 0.0458 to 0.045 organic fraction per minute, targeting a last peak just under 10 minutes. The slightly shallower gradient should improve the late pair and bring the middle pair to approximately baseline resolution.
18	0.15	0.90	14	55	0.932	**0.938**	Test a nearby selectivity balance with a lower starting fraction and a modest temperature reduction. The slope is chosen to keep the last peak near 10 minutes while potentially improving the early cluster and first-pair separation relative to the 60 C run.

#	φ_0	φ_1	t_G	T	CRF	best	the model's rationale, as returned with the proposal
19	0.05	0.70	10	60	0.935	**0.938**	Refine the alternative low-start optimum with a slope just steeper than injection 4, aiming to remove its time penalty while retaining most of its resolution. This tests whether that selectivity pattern can outperform the current best high-start method.
20	0.20	0.90	14	45	0.920	**0.938**	Test whether moving farther beyond the early-pair temperature crossing recovers more resolution than it loses in the late cluster. The steeper gradient compensates for lower temperature, targeting a last peak near 10 minutes.
21	0.05	0.55	4	60	0.803	**0.938**	Test a faster gradient followed by a lower-organic hold to change late-cluster selectivity rather than merely fine-tuning slope. The low starting fraction protects early retention, while 60 C limits the risk of excessive hold-phase retention.
22	0.15	0.90	14	60	0.930	**0.938**	Raise temperature one step from injection 18 to improve the late critical pair while retaining a penalty-free run. This directly tests whether the intermediate starting fraction offers a better overall resolution balance at the highest temperature.
23	0.25	0.95	16	45	0.929	**0.938**	Test whether moderate cooling improves the compressed early and middle pairs at the higher starting fraction. Shortening the gradient relative to injection 9 should offset increased retention, while the late cluster remains the main selectivity risk.
24	0.25	0.75	12	45	0.936	**0.938**	Make injection 23's gradient slightly shallower to spend its remaining time margin on the unresolved middle and late pairs. The last peak is expected just under 10 minutes, potentially improving the CRF without adding a time penalty.
25	0.25	0.90	16	50	0.932	**0.938**	Use a modest temperature increase and slightly shallower slope than injection 24 to improve the late critical pair while targeting a last peak near 10 minutes. This final test balances that potential gain against temperature-driven losses in the early cluster.

▼ LLM search on hplcsim_sb, seed 0, injection by injection

#	φ_0	φ_1	t_G	T	CRF	best	the model's rationale, as returned with the proposal
1	0.05	0.95	12	40	0.887	0.887	Start with a broad gradient to observe all 22 compounds and identify the elution window and critical pairs. A moderate temperature leaves room to probe temperature-dependent selectivity in either direction.
2	0.05	0.95	12	60	0.836	0.887	Keep the gradient fixed and raise temperature to test selectivity changes, particularly in the late co-eluting cluster. Reduced retention may also create time headroom for a shallower gradient if needed.
3	0.05	0.95	12	25	0.905	0.905	Test the opposite temperature limit with the gradient unchanged, since heating worsened the early critical pair without resolving the late co-elution. This will establish whether cooling improves selectivity enough to justify subsequent gradient adjustment.
4	0.05	0.95	8	25	0.852	0.905	Change only gradient duration from the best run to measure slope sensitivity of the unresolved pairs. The steeper gradient also removes the time penalty and may separate the late co-elution if its components have different solvent-strength sensitivities.
5	0.20	0.90	12	25	0.933	0.933	Use a narrower, shallower gradient at the best temperature to improve the slope-sensitive middle and late pairs. Raising the starting organic fraction should offset the added retention time, while testing whether the earliest peaks remain adequately retained.
6	0.30	0.90	12	25	0.923	0.933	Further narrow the gradient to improve the remaining slope-sensitive pairs while advancing the late peaks. The early peaks have enough retention and resolution margin to test a higher starting organic fraction.
7	0.30	0.95	14	25	0.935	**0.935**	Make the gradient modestly shallower than injection 6 while retaining its starting composition. This should improve the middle and late critical pairs while keeping the last peak near, but below, 10 minutes.

#	φ_0	φ_1	t_G	T	CRF	best	the model's rationale, as returned with the proposal
8	0.30	0.65	8	25	0.929	**0.935**	Use a slightly shallower gradient, ending near the late elution window, to improve the remaining critical pairs. The final hold also tests whether changing the late solvent profile can disrupt the persistent co-elution without a substantial time penalty.
9	0.40	0.90	14	25	0.866	**0.935**	Test a substantially higher starting composition with a shallower slope to alter selectivity rather than merely stretch the chromatogram. This may improve the stubborn late cluster, while revealing whether early-pair compression limits further window trimming.
10	0.25	0.85	12	25	0.944	**0.944**	Interpolate between the strongest methods using a moderate starting fraction and a 0.05-per-minute slope. This should preserve early separation while bringing the middle critical pairs closer to full credit with the last peak near 10 minutes.
11	0.30	0.65	8	35	0.928	**0.944**	Revisit injection 8 at a moderately higher temperature to reduce its small time penalty and test selectivity under the shallower gradient. Heating may improve the middle pair while retaining the early-pair resolution gained from the higher starting composition.
12	0.30	0.90	14	25	0.944	**0.944**	Use a slightly shallower slope than injection 8 but a higher endpoint, eliminating its late hold-related delay. This targets full credit for the middle pairs while keeping the last peak close to 10 minutes.
13	0.05	0.65	4	25	0.848	**0.944**	Test a faster gradient followed by a moderate-organic hold, placing the persistent late cluster in a different solvent regime. The low starting fraction protects early retention, while the 0.65 endpoint has already demonstrated complete elution.
14	0.25	0.90	14	30	0.926	**0.944**	Test a slightly shallower gradient near the best method, with modest heating to offset the added retention time. This probes whether improved slope-dependent separation outweighs the small selectivity losses seen on heating.
15	0.30	0.75	10	25	0.941	**0.944**	Refine around injection 12 with a slightly steeper gradient that should put the last peak safely below 10 minutes. This tests whether removing its time penalty preserves enough resolution to improve the best score.
16	0.05	0.60	2	25	0.782	**0.944**	Test a distinct selectivity regime: a rapid gradient followed by a lower-organic hold, so more of the mixture separates during the hold. This may break the persistent late co-elution that has barely responded to temperature or shallow-gradient adjustments.
17	0.35	0.95	16	25	0.922	**0.944**	Test the remaining intermediate starting composition with a shallower gradient designed to keep the last peak near 10 minutes. This checks whether gains in the late cluster can outweigh compression of the early and middle critical pairs.
18	0.30	0.90	14	30	0.939	**0.944**	Raise temperature just one step from injection 12 to remove its small time penalty. Keeping the gradient unchanged will test whether the resulting selectivity trade-off improves the score near the current optimum.
19	0.20	0.75	10	25	0.933	**0.944**	Test a slope between injections 5 and 10 at the lower starting composition, which better preserves the middle-pair selectivity. The endpoint remains high enough to avoid the delayed final hold seen with 0.65 organic.
20	0.25	0.85	12	30	0.939	**0.944**	Test a single temperature step above the best method while holding its gradient fixed. This isolates whether modest selectivity changes can improve the score without introducing a time penalty.
21	0.15	0.90	12	25	0.928	**0.944**	Test a lower starting fraction with a compensating steeper slope, targeting a last peak near 10 minutes. This checks whether improved middle-pair selectivity can outperform the higher-starting-fraction methods.
22	0.30	0.80	12	25	0.936	**0.944**	Test a slightly shallower slope than injection 12 to improve the remaining middle and late pairs. The expected small time penalty will establish whether additional resolution can outweigh crossing the 10-minute boundary.
23	0.30	0.80	12	30	0.939	**0.944**	Raise injection 22 by one temperature step to recover most of its time penalty while retaining its shallower slope. This tests the remaining local trade-off between improved resolution and the 10-minute cutoff.

#	φ_0	φ_1	t_G	T	CRF	best	the model's rationale, as returned with the proposal
24	0.20	0.85	12	35	0.932	**0.944**	Test a slightly shallower gradient at the lower starting fraction, using moderate heating to keep the last peak near 10 minutes. This targets the favorable middle-pair separation seen in injection 19 while reducing its time penalty.
25	0.30	0.65	8	30	0.938	**0.944**	Test the intermediate temperature between injections 8 and 11, aiming to remove the late-hold time penalty while limiting the selectivity losses observed at 35 C. This is a final local refinement near the strongest gradient slopes.

What carries over, and what does not

The ranking of the three searches is the same as on the generated sets, and the LLM search's advantage is if anything larger here, because the good region is smaller and sits at a temperature bound that a one-line reading of the first chromatogram points to. Two cautions apply. Each method ran ten campaigns per set here, not the fifty of section 8, so a one-in-ten event like BO's single success is measured coarsely. And the sets are real compounds seen through the simulator's model: the fit is close for 41 of 43 compounds, but the two weakly retained ones carry the model's inability to let the slope drift with temperature, and the instrument, the peak-width model and the 42 min window are still the simulator's. Files: `hplcsim/configs/analytes/hplcsim_*.yaml` and `.import.json`, `hplcsim/tables/`, `hplcsim/results/`; rebuild with `python -m sim.import_hplcsim --table`.

10. The prompt taken apart

Section 5 lists what the LLM search is told that BO is not: the scoring rule, linear-solvent-strength theory and a few rules of method-development practice, all in the system prompt. The warm start of section 6 showed that the advantage survives when the observations are equalised; this section asks how much of the prompt it needs. The first deletion is the retention-model section (the 21 lines headed "Retention model" in section 5), with the same model (`gpt-6-astra`), harness, grid and budget and 50 campaigns per set, and nothing else changed.

variant	set	found optimum	injections to optimum, median (IQR; max)	found by 5 / 10 / 15	truncated injections	CRF of first injection
LLM, full prompt (section 8)	easy	50 of 50	1 (1–1; 1)	50 / 50 / 50	0	1.000
	medium	50 of 50	6 (5–6; 10)	15 / 50 / 50	0	0.592
	hard	50 of 50	4 (3–4; 5)	50 / 50 / 50	0	0.931
LLM, prompt without the retention model	easy	50 of 50	1 (1–1; 1)	50 / 50 / 50	0	1.000
	medium	50 of 50	6 (5–8; 10)	13 / 50 / 50	0	0.599
	hard	50 of 50	4 (3–4; 6)	48 / 50 / 50	0	0.934

gpt-6-astra, 50 campaigns of 25 injections per row. "Found" is a best CRF within 1 % of the optimum; the injection statistics are over the campaigns that found it. No campaign in either variant ran a truncated method or repeated a proposal.

Without the retention model: nothing changes

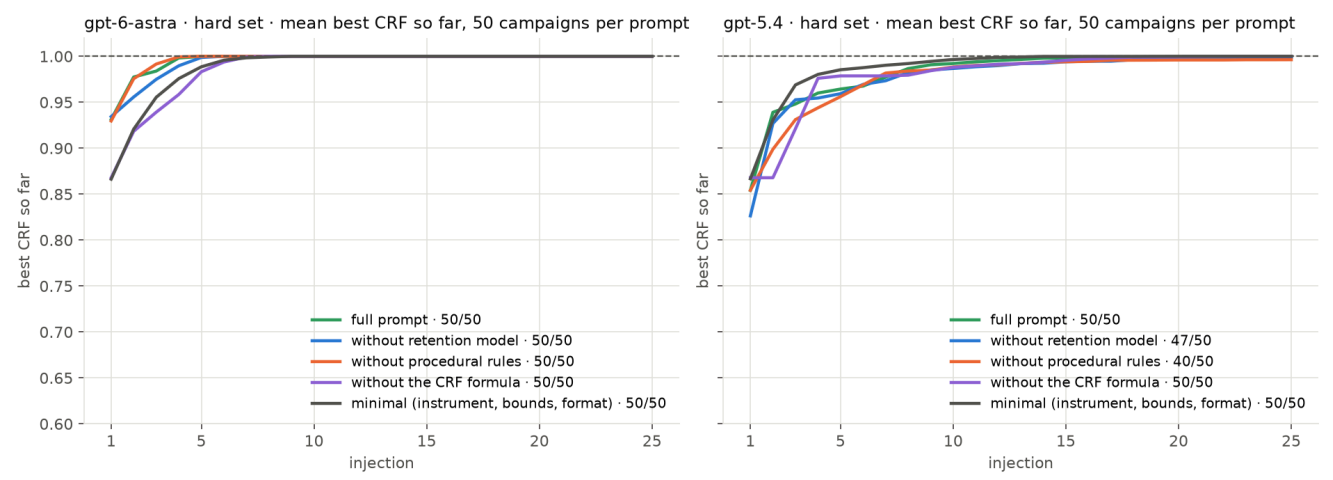
Deleting the 21 lines of LSS theory from the prompt changed nothing measurable. All 150 campaigns reached the optimum after the same medians (1, 6 and 4 injections), the same first-injection CRF (1.000, 0.599 and 0.934 against 1.000, 0.592 and 0.931), no truncated method and no repeated proposal. The rationales read the same: in both variants the model writes about slope in about a third of its rationales (31 % against 37 %), temperature in about 40 % and selectivity in about a quarter, and in neither does it ever cite log k, S or the LSS equation. The retention theory the search uses is the model's own, not the prompt's. What the prompt supplied and was not ablated here is the scoring rule, the bounds and the instruction to spend early injections on information; the five nested prompts below take those apart.

Five nested prompts on the hard set

That removed one section of the prompt on three sets. To see how much of the prompt the search needs at all, and whether the answer depends on the model, five nested versions were run on the hard set with `gpt-6-astra` and with the second model of section 11, `gpt-5.4` , 50 campaigns each: the full prompt of section 5; without the retention-model section, as above; without the four procedural rules under "How to work" (one grid method per injection and no repeats, early injections buy information, resolve every pair then trade time, avoid truncation), with the theory kept; without the CRF formula, which is replaced by "higher is better" (the procedural rule that restates it is removed with it); and a minimal prompt that keeps only the instrument, the method bounds, the format of a run and the answer format. Each variant is generated from the full prompt by deleting or replacing whole sections (`ablations/prompts/make_variants.py`), so the files differ only where stated. In every variant the model still receives the CRF and its two terms after each injection, and the harness still refuses a repeated method without charging it.

model	prompt	found optimum	injections to optimum, median (IQR; max)	found by 5 / 10 / 15	CRF of first injection	opening method (campaigns)	re-asks per campaign
gpt-6-astra	full	**50 of 50**	4 (3–4; 5)	50 / 50 / 50	0.931	0.05 → 0.95 / 10 min / 40 °C (31)	0.00
	without retention model	**50 of 50**	4 (3–4; 6)	48 / 50 / 50	0.934	0.05 → 0.95 / 10 min / 40 °C (16)	0.00
	without procedural rules	**50 of 50**	3 (3–4; 4)	50 / 50 / 50	0.930	0.05 → 0.95 / 10 min / 40 °C (32)	1.00
	without the CRF formula	**50 of 50**	5 (5–6; 8)	30 / 50 / 50	0.868	0.05 → 0.95 / 20 min / 40 °C (50)	0.04
	minimal	**50 of 50**	5 (4–6; 9)	33 / 50 / 50	0.866	0.05 → 0.95 / 20 min / 40 °C (47)	1.02
gpt-5.4	full	**50 of 50**	10.5 (7–13; 21)	8 / 25 / 44	0.854	0.05 → 0.95 / 20 min / 40 °C (49)	0.28
	without retention model	47 of 50	9 (6–11; 14)	9 / 32 / 47	0.826	0.05 → 0.95 / 20 min / 40 °C (39)	0.26
	without procedural rules	40 of 50	10 (7–14; 23)	7 / 22 / 34	0.854	0.05 → 0.95 / 20 min / 40 °C (49)	1.48
	without the CRF formula	**50 of 50**	14 (13–15; 22)	2 / 4 / 38	0.868	0.05 → 0.95 / 20 min / 40 °C (50)	0.38
	minimal	**50 of 50**	8 (4–10; 14)	18 / 40 / 50	0.867	0.05 → 0.95 / 20 min / 40 °C (28)	1.26

Hard set, 50 campaigns of 25 injections per row. The opening method is the most common first injection with the number of campaigns that used it. Re-asks are proposals of an already-run method, refused by the harness and asked again without charge. No variant ran a truncated method with gpt-6-astra; gpt-5.4 ran one or two in 1 250 injections with the full and the no-theory prompt.



Mean best CRF so far against injection for the five prompt variants on the hard set, 50 campaigns each, with gpt-6-astra (left) and gpt-5.4 (right). The legend gives the campaigns that reached the optimum. File: `ablations/prompts/prompt_study.png` .

gpt-6-astra: the prompt changes where the search starts, not where it ends

Every prompt reached the optimum in 50 of 50 campaigns, after a median of 4, 4, 3, 5 and 5 injections and never more than 9. Told the formula, the model opens with a 10 min full-window scouting gradient in 31 of 50 campaigns and a first-injection CRF of 0.93; without the formula it opens with a 20 min gradient in 47 to 50 campaigns (0.87), reads from the returned time term that the last peak costs, and shortens t_G from its second injection. The cost is one injection in the median and a drop from 50 to about 30 campaigns at the optimum by injection 5; by injection 10 all five variants are at 50. Removing the procedural rules removes the instruction not to repeat a method, and the model then proposed an already-run method about once per campaign (1.0 re-asks per campaign against 0 with the rules). The harness refuses such a proposal and asks again for free; without that rule in the

harness, the no-rules and minimal prompts would have spent about one injection per campaign on repeats.

> None of the domain content in the prompt, whether retention theory, procedure or the scoring formula, is needed by gpt-6-astra to reach the optimum of the hard set within the budget. What the formula buys is a short opening gradient, worth one or two injections at the start, and what the procedural rules buy is the no-repeat discipline the harness enforces anyway. The retention theory the search uses, and the reading of the table that the warm start isolated, are the model's own. Whether that holds for a different model is the question of section 11. Files: `ablations/no_theory/`, `ablations/prompts/<variant>/<model>/`; summarise with `python ablations/summary.py` and `python ablations/prompts/prompt_summary.py`.

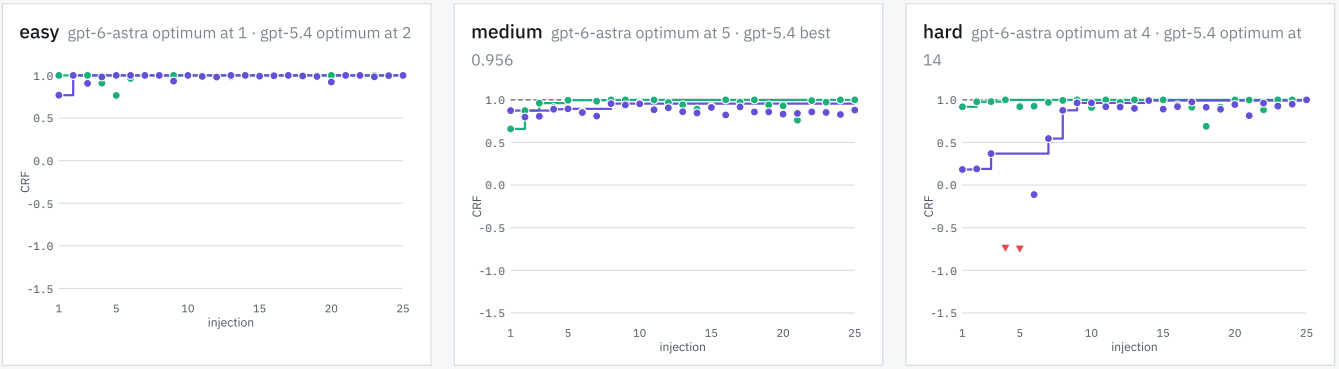
11. The same search on a second model

Sections 5 to 10 use one model, `gpt-6-astra`. That leaves open how much of the LLM search's result is the model and how much is the prompt and harness around it. This section repeats the whole LLM benchmark with a second Azure OpenAI deployment, `gpt-5.4`, and changes nothing else: the same system prompt, the same per-injection user message, the same grid, snap and repeat guard, the same three generated sets, the same budget of 25 and the same 50 campaigns per set. Only the deployment name in the environment file differs, and the second model's campaigns are kept in their own results folder so the resumable runner could not pick up the first model's transcripts.

The figures and tables below put the two models side by side in the format of sections 7 and 8: one seed-0 campaign per set, then the 50-campaign averages, the checkpoint table and the cost of the calls.

One campaign per set

Seed 0 of each model on each set: the CRF of every injection as a dot, the best so far as a step line, a truncated run as a red triangle. The seed only labels the repeat; both models start every campaign from a blank page.



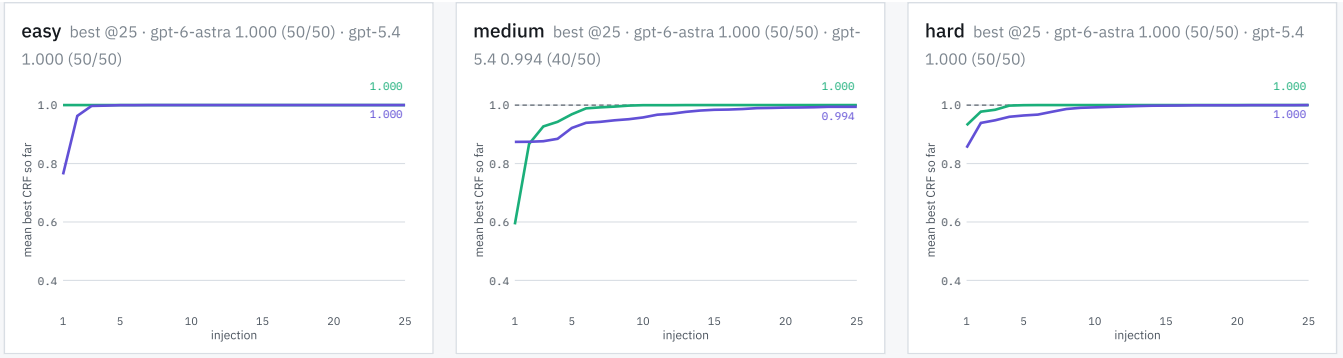
Seed 0, 25 injections. First injection at the optimum, easy: gpt-6-astra at injection 1, gpt-5.4 at injection 2; medium: gpt-6-astra at injection 5, gpt-5.4 not within 25 (best 0.956); hard: gpt-6-astra at injection 4, gpt-5.4 at injection 14. gpt-5.4 runs 2 truncated methods across the three campaigns; gpt-6-astra runs none.

Fifty campaigns per set

The same statistics as section 8, for the two models. "Best @25" is the best CRF a campaign had after its 25th injection, averaged over the 50 repeats; "found optimum" counts the campaigns whose best was within 1 % of the optimum; "injections to optimum" is the median over the campaigns that found it, with the range in brackets.

> gpt-5.4 reached the optimum in **50 of 50** campaigns on easy, **40 of 50** on medium and **50 of 50** on hard, against 50, 50 and 50 of 50 for gpt-6-astra. Median injections to the optimum, gpt-5.4 against gpt-6-astra: easy 3 vs 1, medium 12 vs 6, hard 10.5 vs 4.

Set	Model	Mean best @25	Found optimum	Injections to optimum, median (range)	Worst campaign
easy	gpt-6-astra	**1.000**	**50/50** (100 %)	1 (1 to 1)	1.000
easy	gpt-5.4	**1.000**	**50/50** (100 %)	3 (2 to 6)	1.000
medium	gpt-6-astra	**1.000**	**50/50** (100 %)	6 (4 to 10)	1.000
medium	gpt-5.4	**0.994**	**40/50** (80 %)	12 (4 to 23)	0.944
hard	gpt-6-astra	**1.000**	**50/50** (100 %)	4 (2 to 5)	1.000
hard	gpt-5.4	**1.000**	**50/50** (100 %)	10.5 (2 to 21)	1.000



Mean best CRF so far against injection number, 50 campaigns per model and set, on the axes of the section 8 figure. The gap between the curves is the cost of the model choice in injections: where gpt-6-astra's curve has already flattened at 1.0, gpt-5.4's is still climbing.

set	model	mean best @5 (share at optimum)	mean best @10 (share at optimum)	mean best @15 (share at optimum)	mean best @20 (share at optimum)	mean best @25 (share at optimum)
easy	gpt-6-astra	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)
easy	gpt-5.4	1.000 (98 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)
medium	gpt-6-astra	0.968 (30 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)
medium	gpt-5.4	0.922 (4 %)	0.958 (20 %)	0.984 (68 %)	0.992 (78 %)	0.994 (80 %)
hard	gpt-6-astra	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)	1.000 (100 %)
hard	gpt-5.4	0.964 (16 %)	0.992 (50 %)	0.998 (88 %)	0.999 (98 %)	1.000 (100 %)

What the calls cost

Per campaign of 25 calls. Prompt tokens are nearly identical because the conversation is the same append-only transcript; the difference is in how much each model writes back, how long each call takes and how often it proposed a method that had already been run (which the harness refuses and re-asks without charging an injection).

set	model	campaigns	prompt tokens per campaign	completion tokens per campaign	mean latency per call (s)	proposals re-asked (repeat of a run method)	unparseable replies
easy	gpt-6-astra	50	268,644	2,906	4.6	0	0
easy	gpt-5.4	50	273,967	2,492	3.0	1	0
medium	gpt-6-astra	50	283,667	6,620	8.7	0	0
medium	gpt-5.4	50	290,392	2,549	3.2	45	0
hard	gpt-6-astra	50	295,053	6,644	9.3	0	0
hard	gpt-5.4	50	300,296	2,558	3.0	12	0

Reading the result

On the final score the two models agree on two of the three sets. On easy and hard every campaign of either model ends within 1 % of the optimum, so on those sets the ceiling is set by the prompt and harness, not by the model. Medium is where they part: gpt-5.4 leaves 10 of 50 campaigns short, with a worst final CRF of 0.944, where gpt-6-astra reaches the optimum every time. Medium is also the set that gpt-6-astra found slowest in section 8, so it is the one that discriminates between models, despite its name.

On sample efficiency the models differ everywhere. gpt-6-astra is at the optimum by a median of 1, 6 and 4 injections on easy, medium and hard; gpt-5.4 needs 3, 12 and 10.5, with a far wider spread (up to 21 on hard and 23 on medium). The checkpoint table shows the same thing as a share: after five injections 100 % of gpt-6-astra's hard campaigns are at the optimum against 16 % of gpt-5.4's, and the gap only closes at injection 20.

The cost table points at one reason. gpt-5.4 proposed a method that had already been run 58 times over the 150 campaigns (45 of them on medium), against 0 for gpt-6-astra; each is a sign of losing track of the transcript it is reading, and the harness only refuses the repeat, it does not say what to do instead. gpt-5.4 also writes about 2600 completion tokens per campaign on hard against about 6600 for gpt-6-astra, and answers in about a third of the time: its proposals are shorter and quicker, and on the sets with a narrow good region, less well aimed.

So the answer to the question this section asks is in two parts. That the LLM search beats random search and BO (section 8) is a property of the method: both models do, by a wide margin, on every set. How fast it gets there, and whether it gets there at all on the set with the smallest good region, depends on the model. A reader who wants to attribute the benchmark result to the prompt rather than to the model should expect the margin over BO to survive a change of model and the sample efficiency not to.

The five prompts with gpt-5.4: the same in kind, noisier in degree

The five prompts of section 10 were also run on the hard set with gpt-5.4 (second half of the table and the right panel of the figure there). The full prompt, the prompt without theory and the prompt without the formula reached the optimum in 50, 47 and 50 campaigns after a median of 10.5, 9 and 14 injections. The minimal prompt was its fastest at 8 (interquartile range 4 to 10, at most 14; Mann–Whitney against the full prompt, $p < 0.001$), and the prompt without procedural rules its slowest, at the optimum in 40 campaigns of 50 (Fisher's exact test against 50 of 50, $p < 0.001$), with the other ten ending at a best CRF of 0.965 to 0.99, one grid step short. gpt-5.4 also re-proposed run methods more often without the rule (1.3 to 1.5 re-asks per campaign against 0.3 with it). The order of the variants is not the same for the two models, and the spread between prompts on gpt-5.4 (medians 8 to 14) is smaller than the spread between the models on the same prompt (4 against 10.5).

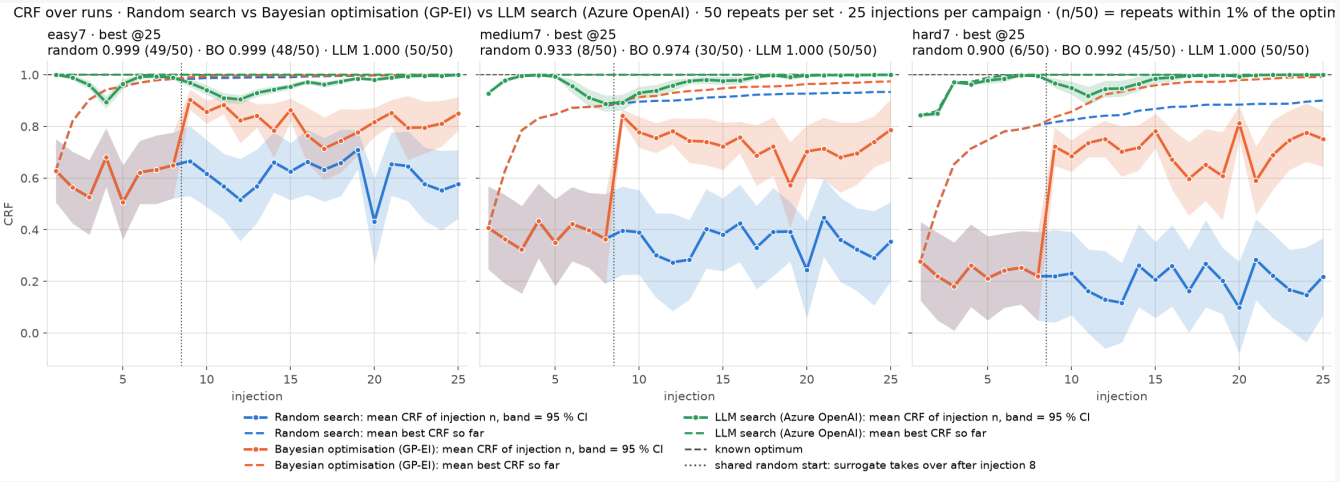
> Two things follow. Neither model needs the domain content of the prompt to reach the optimum of the hard set within the budget, as section 10 found for the first. And the prompt is a second-order effect next to the model: the same five prompts span one injection of median with gpt-6-astra and six with gpt-5.4, and no prompt brings gpt-5.4 to gpt-6-astra's count. The study is confined to the hard set, and 50 campaigns resolve differences of about ten in the found count and about two injections in the median; smaller effects are within noise. Files: `ablations/prompts/<variant>/<model>/` ; summarise with `python ablations/prompts/prompt_summary.py` .

12. Fresh mixtures: the result is not three sets

Everything so far rests on three synthetic mixtures and two measured ones. To check that the result is not a property of those three draws, the unchanged prompt and code were applied to three mixtures generated afresh from the same generator with seed 7 (easy7, medium7 and hard7, with 6, 8 and 10 analytes), made after the prompt was frozen, with random search and BO rerun on them for comparison. Same model, harness, grid and budget, 50 campaigns per search and set.

variant	set	found optimum	injections to optimum, median (IQR; max)	found by 5 / 10 / 15	truncated injections	CRF of first injection
Random search, fresh family	easy7	49 of 50	4 (2–8; 22)	29 / 38 / 42	6.0 %	0.629
	medium7	8 of 50	9.5 (8–18; 21)	2 / 5 / 5	12.9 %	0.408
	hard7	6 of 50	10 (8–16; 18)	1 / 3 / 4	15.6 %	0.278
BO, fresh family	easy7	48 of 50	4 (2–8; 23)	29 / 44 / 45	2.9 %	0.629
	medium7	30 of 50	17.5 (12–20; 25)	2 / 7 / 13	6.0 %	0.408
	hard7	45 of 50	16 (14–20; 25)	1 / 3 / 21	7.0 %	0.278
LLM, fresh family (seed 7)	easy7	**50 of 50**	1 (1–1; 1)	50 / 50 / 50	0	1.000
	medium7	**50 of 50**	2 (2–3; 3)	50 / 50 / 50	0	0.929
	hard7	**50 of 50**	6 (5–6; 7)	22 / 50 / 50	0	0.843

50 campaigns of 25 injections per row. "Found" is a best CRF within 1 % of the optimum; the injection statistics are over the campaigns that found it; "truncated" is the share of injections that ran a truncated method. Random search and BO share injections 1 to 8 for a given seed. medium7 and hard7 have 118 and 112 grid rows within 1 % of the optimum (against 500 and 82 for medium and hard); easy7 has 2 929, 13 % of the grid.



The fresh family: mean CRF of each injection (dots, with the 95 % band) and mean best CRF so far (dashed) over 50 campaigns per search and set, on the same axes as the section 8 figure. Random search and BO share injections 1 to 8. File: `ablations/fresh/results/crf_over_runs.png` .

Nothing in the prompt, code or grid was changed between the original family and this one, and the three sets were generated after the prompt was frozen. The ranking and the margins reproduce: the LLM reached the optimum in all 150 campaigns after a median of 1, 2 and 6 injections; BO in 48, 30 and 45 after 4, 17.5 and 16; random search in 49, 8 and 6. So the result on easy, medium and hard is not a property of those three mixtures. The order of difficulty is also not the one BO sees: medium7, where BO found the optimum in 30 campaigns of 50, took the LLM two injections, while hard7, where BO found it in 45, took six. The hard7 optimum starts the gradient at $\varphi_0 \approx 0.13$ rather than at the lower bound of the grid, and the LLM's full-window first injection scored only 0.84 there against 0.93 on medium7. Across the 600 gpt-6-astra campaigns of sections 6, 10 and 12 and the 150 of section 8 the LLM search has not run a truncated method.

> Taken together, sections 6, 10, 11 and 12 answer the four checks section 5 named. The advantage is not where the search starts, since BO's start can be imposed without changing the outcome; it is not an artefact of the prompt's theory, since the theory can be deleted; it is not memorisation of three mixtures; and it holds on a second model, with a larger budget. What they leave open is the comparison with retention modelling (a two-gradient LSS fit as a fourth baseline), which no ablation of the LLM can answer, and the step from a noise-free simulator to an instrument. Files: `ablations/fresh/` ; summarise with `python ablations/summary.py` .